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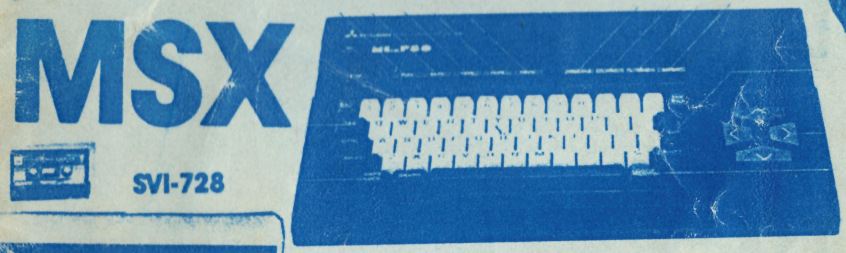
SOFTWARE

MSX



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SVI-728

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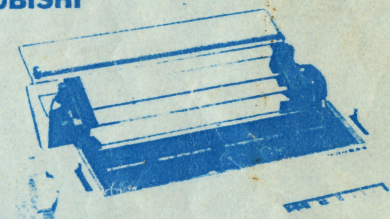
SVI 328



brother

queries

SOFTWARE REVIEW

CASSETTE LOADING PROBLEMS.

IT SEEMS MORE AND MORE PEOPLE ARE HAVING PROBLEMS WITH THE LOADING OF PROGRAMMES FROM CASSETTE.
THIS APPLIES MORE TO THE '767' FOR MSX-USE THAN TO THE '904' FOR 318/328-USE BUT THE PRINCIPLES APPLY EQUALLY TO ALL DATA-RECORDERS.

THE PROBLEMS CAN BE CAUSED BY A VARIETY OF FACTORS, OR EVEN TO A COMBINATION OF ONE OR MORE OF THE FACTORS AS OUTLINED BELOW.....

1)...DIRTY DATA RECORDER.
USING A COTTON-BUD AND METHS, GENTLY CLEAN THE ERASE-HEAD, PLASTIC AUTO-STOP LEVER, RECORDING HEAD AND BLACK RUBBER-WHEEL. THE PLAY-BUTTON SHOULD BE DEPRESSED TO BRING THE CARRIAGE FORWARD. SHAKE EXCESS METHS OFF FIRST.

2)...PROXIMITY OF OTHER EQUIPMENT.
ENSURE THAT THE DATA-RECORDER IS WELL CLEAR OF THE MONITOR, COMPUTER AND POWER-SUPPLY. STRAY MAGNETIC FIELDS CAN CAUSE A BAD OR NO-LOAD.

3)...MAINS VOLTAGE FLUCTUATIONS.
EVEN THOUGH THE DATA-RECORDER CAN TOLERATE AN INCREASE OR DECREASE IN VOLTAGE OF +- 5%, THE MAINS-VOLTAGE MAY VARY EVEN MORE THAN THIS. IF A RECORDING IS MADE AT 220V AND PLAYBACK IS AT 205V, THE TAPE MAY BE TOO SLOW. SIMILARLY, IN PRETORIA FOR EXAMPLE, WHERE THE MAINS-VOLTAGE, IS 250V, THE TAPE-SPEED MAY BE TOO HIGH. THIS DOES NOT OF COURSE, APPLY TO THE '904', AS THE VOLTAGE IS REGULATED IN THE 318/328.

4)...TAPE-MODULATION.
IF THE LEVEL OF VOLUME ON THE TAPE IS TOO LOW, THE OUTPUT FROM THE AMPLIFIER MAY BE TOO LOW. SIMILARLY, TOO HIGH A TAPE-MODULATION CAN ALSO CAUSE DISTORTION IN THE AMPLIFIER (OVER-LOAD).

5)...FAULTY TAPE.
THIS IS THE LEAST-COMMON CAUSE OF BAD LOADING. IF A DEVICE-I/O-ERROR OCCURS, REMOVE THE TAPE, WITHOUT REWINDING AND LOOK AT THE TAPE. IT MAY BE, THAT THERE IS A CREASE ON THE TAPE. IT IS ALSO POSSIBLE THAT THE TAPE HAS BEEN LEFT TOO CLOSE TO A SPEAKER, TRANSFORMER, MONITOR, ETC. THIS CAUSES 'MAGNETIC-WIPING' OF THE DATA ON A TAPE. HEAT COULD ALSO BE FATAL TO A TAPE.

6)...FAULTY DATA-RECORDER.
DROPPING THE RECORDER CAN CAUSE DAMAGE. DURING TRANSIT OR IN THE HOME, THE HEAD MAY BE CAUSED TO BECOME MISALLIGNED. EVEN 'HEAVY-HANDEDNESS' ON THE BUTTONS CAN CAUSE CARRIAGE PROBLEMS.

7)...BAD CONNECTIONS.
THE CONNECTOR TO THE COMPUTER MAY BE FAULTY IN THAT THE PINS MAY BE CORRODED OR LOOSE.

8)...BUMPING.
BUMPING THE RECORDER DURING SAVING OR LOADING, COULD MOVE THE HEAD SLIGHTLY. THE RECORDER CAN BE PLACED LEVEL ON A PIECE OF POLYSTYRENE FOAM AND WELL CLEAR OF YOUR 'ELBOW'.

MSX...MSX...MSX...MSX
ONLY

```
10 '************************
20 '*      M.S.X PACMAN      *
30 '*     by C.R.WALTERS     *
40 '* SPECTRAVIDEO SVI 728 *
50 '************************
60 CLEAR500
70 ONSTOPGOSUB730:STOPON
80 DEFINTA-Z
90 DEFUSR0=&H41:DEFUSR1=&H44
100 N=RND(-TIME):CL=8192:GOSUB670
110 SCREEN1,0:WIDTH32:KEYOFF
120 GOSUB610:GOSUB400
130 CLS:LI=3:GOSUB460
140 U=USR1(0):SC=0:TI=800:LOCATE1,22:PRI
NT"LIVES";LI;SPC(2)"SCORE";SC:LOCATE21,2
2:PRINT"TIME";TI
150 Z=6799:VPOKEZ,96
160 R(0)=INT(RND(1)*19)+1:C(0)=INT(RND(1
)*13)+2:IFVPEEK(32*R(0)+C(0)+6144)<>46TH
EN160
170 R(1)=INT(RND(1)*19)+1:C(1)=INT(RND(1
)*13)+16:IFVPEEK(32*R(1)+C(1)+6144)<>46T
HEN170
180 E(0)=8:E(1)=13:FORB=0TO1:PUTSPRITEB,
(C(B)*8,R(B)*8-1),E(B),0:PUTSPRITE2,(C(B
)*8,R(B)*8-1),7,1:NEXTB
190 S=STICK(0):IFS=0THEN240
200 X=32*(S=1)-(S=3)-32*(S=5)+(S=7)
210 W=VPEEK(Z+X):IFW=120THEN240
220 IFW=46THENSC=SC+1:LOCATE16,22:PRINTS
C:IFSC=234THEN590
230 VPOKEZ,32:Z=Z+X:VPOKEZ,96
240 FORD=0TO1
250 IFVPEEK(32*R(D)+C(D)+6144)=96THENY1=
0:X1=0:GOTO350
260 TI=TI-1:LOCATE25,22:PRINTTI:IFTI=0TH
EN360
270 A=INT(RND(1)*4)+1:ONAGOTO280,290,300
,310
280 X1=1:Y1=0:GOTO320
290 Y1=-1:X1=0:GOTO320
300 X1=-1:Y1=0:GOTO320
310 Y1=1:X1=0
320 P=VPEEK(32*(R(D)+Y1)+C(D)+X1+6144):I
FP=96THEN350ELSEIFP=120THEN340
330 R(D)=R(D)+Y1:C(D)=C(D)+X1:PUTSPRITED
,(C(D)*8,R(D)*8-1):PUTSPRITE2,(C(D)*8,R(
D)*8-1)
340 NEXTD:GOTO190
350 R(D)=R(D)+Y1:C(D)=C(D)+X1:PUTSPRITED
,(C(D)*8,R(D)*8-1):PUTSPRITE2,(-24,0)
360 GOSUB660:FORJ=1TO8:VPOKECL+2,160:FOR
DL=1TO30:NEXTDL:VPOKECL+12,96:FORDL=1TO3
0:NEXTDL:VPOKECL+12,64:FORDL=1TO30:NEXTD
L,J
370 VPOKEZ,32:LI=LI-1:FORDL=1TO250:NEXTD
L:LOCATE6,22:PRINTLI
```

```
380 FORDL=1TO500:NEXTDL:FORQ=0TO2:PUTSPR
ITE0,(-24,0):NEXTQ:IFLI=0THEN560
390 CLS:GOSUB460:GOTO140
400 FORI=1TO3:M$="":C=0:READM,M$
410 FORJ=1TOLEN(M$)STEP2
420 VPOKEM*8+C,VAL("&H"+MID$(M$,J,2)):C=
C+1:NEXTJ,I
430 DATA 46,0000001818000000,69,183C183C
5A182466,120,DFDF00FBFBFB00DF
440 M$(0)="66247EDBBDFFFF99":M$(1)="0000
002424000000"
450 FORI=0TO1:S$="":FORJ=1TOLEN(M$(I))ST
EP2:S$=S$+CHR$(VAL("&H"+MID$(M$(I),J,2))
):NEXTJ:SPRITE$(I)=S$:NEXTI:RETURN
460 U=USR0(0):COLOR1,15,15:VPOKECL+11,96
:VPOKECL+12,64:VPOKECL+15,&HCA
470 PRINT" xxxxxxxxxxxxxxxxxxxxxxxxxxxxx
   x............................x"
480 PRINT" x.x.xxxxxxxxxx.xxxxxxxxxx.x.x
   x............x.x............x"
490 PRINT" x.xxxxxxxxxx.x.x.xxxxxxxxxx.x
   x.xxxxxxxxxx.x.x.xxxxxxxxxx.x"
500 PRINT" x.xxxxxxxxxx.x.x.xxxxxxxxxx.x
   x............x.x.............x   x.x.x
xx.xxxx.x.x.xxxx.xxx.x.x"
510 PRINT" x......xxxx.x.x.xxxx.......x
   x.x.xxx.xxxx.x.x.xxxx.xxx.x.x   x.x.x
xx.xxxx.x.x.xxxx.xxx.x.x"
520 PRINT" x.x.xxx......x.x......xxx.x.x
   x.x.xxxxxxxx.x.x.xxxxxxxx.x.x   x.x.x
xxxxxxx.x.x.xxxxxxxx.x.x"
530 PRINT" x.x.xxxxxxxx.x.x.xxxxxxxx.x.x
   x.x.....................x.x   x.xxx
xxxxxxxx.xxxxxxxxxxx.x"
540 PRINT" x.........................x
   xxxxxxxxxxxxxx xxxxxxxxxxxxxx"
550 RETURN
560 COLOR1,11,11:CLS:LOCATE9,5:PRINT"ALL
 LIVES LOST"
570 LOCATE2,18:PRINT"PRESS SPACE-BAR TO
PLAY AGAIN"
580 IFNOTSTRIG(0)THEN580ELSE130
590 FORL=0TO2:PUTSPRITEL,(-24,0):NEXTL:C
OLOR4,14,6:CLS
600 LOCATE1,5:PRINT"WELL DONE, YOU CLEAR
ED THE MAZE":GOTO570
610 CLS:COLOR1,11,11
620 PRINTSPC(9);"INSTRUCTIONS";SPC(9);"*
**********";:PRINT:PRINT:PRINT:PRINTSPC
(6)"USING THE CURSOR KEYS":PRINT
630 '
640 PRINT"AND GHOSTS WILL TRY TO STOP YO
U.":PRINT:PRINT:PRINT"   PRESS SPACE-BAR
 TO PLAY"
650 IFNOTSTRIG(0)THEN650ELSECOLOR14,14,1
4:RETURN
660 PLAY"T64L6406CEGO5CEGO4CEGO3CEG":RET
URN
670 COLOR15,15,15:SCREEN3
680 OPEN"GRP:"AS#1
690 FORL=1TO10:T$(0)="M.S.X":A(0)=54:B(0
)=24
700 T$(1)="PAC-MAN":A(1)=20:B(1)=120
710 FORV=0TO1:FORT=1TOLEN(T$(V)):COLORRN
D(1)*12+2:PRESET(A(V),B(V)):PRINT#1,MID$
(T$(V),T,1):A(V)=A(V)+32:NEXTT,V,L
720 RETURN
730 SCREEN1:COLOR15,4,4
```

```
**********************

NEXT MONTH...HI-LO

**********************
```

```
THIS PROGRAM WAS TESTED AND RUN
ON AN SVI728 AND PRINTED OUT ON
A TAXAN KP-810 PRINTER USING 40
COLUMN WIDTH IN WIDE-PRINT  AND
DOUBLE-STRIKE MODE.
NOTE..A <ZERO> LOOKS LIKE....0
......A LETTER O LOOKS LIKE..O
```

```
**********************
```

A deaf and dumb man went into an ironmonger's shop to buy a hammer. He couldn't speak so he mimed hammering... then another deaf and dumb man went in wanting a screwdriver, so he mimed putting in a screw...Then a blind man went into the shop and he wanted a saw. So what sort of mime did he do?

He didn't mime at all — he just asked for it!

YOU CAN HIDE THE SCREEN WHILE A PICTURE IS BUILT

SCREEN DISABLE

Q. Is there a way to hide the screen whilst a picture is built?

A. Yes the screen can be disabled whilst a picture is drawn and then when the screen is enabled the picture is displayed instantly. Your program should be written in the following general form:

1) Set SCREEN mode (eg. SCREEN 1 SV1318/328 or SCREEN 2 MSX)
2) Disable screen.
3) Basic instructions to construct picture.
4) Enable screen.

Disable/Enable the screen as follows:

MSX

DEFUSR=&H0041:
Z=USR(0) - disable
DEFUSR=&H0044:
Z=USR(0) - enable

SVI318/328

X=PEEK(&HFA07):
OUT&H81,(X AND &HBF):
OUT&H81,&H81 - disable
OUT&H81,X:OUT&H81,
&H81 - enable

```
1 REM ****************************
2 REM *       SV.328 ONLY        *
3 REM *       ***********        *
4 REM *  INTRODUCTION TO A GAME   *
5 REM *      BY F.C.P. VENTER     *
6 REM ****************************
10 CLEAR 200,&HA000
15 ONSTOPGOSUB1000:STOPON
20 SCREEN1
30 COLOR15,1,5
40 DEFUSR1=&H4782:'   SV LOGO
50 Z=USR1(0)
60 POKE&HFE3A,1
70 LOCATE105,120:PRINT"PRESENTS"
80 LOCATE90,135:PRINT"BATTLE STAR"
90 LOCATE190,160:PRINT"BY ICE"
100 FORT=1TO3000:NEXT
110 COLOR15,4,5
120 SCREEN0,0:LOCATE,,0
130 POKE&HFE35,1:LOCATE17,3:PRINT" MENU"
140 POKE&HFE35,0:LOCATE7,7:PRINT"1  PRAC
TICE      /ONE PLAYER"
150 LOCATE7,9:PRINT"2  NORMAL PLAY   /ON
E PLAYER"
155 LOCATE7,11:PRINT"3  ARCADE LEVEL  /O
NE PLAYER"
160 GOTO160
1000 COLOR15,4,5:SCREEN0,1:LOCATE,,1
```

DIE GEBRUIK VAN DIE LUS IN REKENAARPROGRAMME

deur

H.J. LE ROUX.

DEEL DRIE.....

****

Om die verskil in tyd wat in beslag geneem word deur heeltalle, enkelnoukeurige en dubbelnoukeurige getalle te bepaal, kan die volgende program gebruik word:

```
10 DEFDBL A  of DEFSNG A, of DEFINT A
20 TIME=0
30 FOR A = 1 TO 200 : NEXT
40 PRINT TIME
```

Hieruit sal dit vasgestel word dat dubbelnoukeurige getalle byna twee keer soveel tyd in beslag neem as heeltalle.

In programme waar uitvoeringstyd van deurslaggewende belang is, of wanneer geheuespasie beperk is, is die gebruik van heeltalle beslis 'n belangrike besparingsfaktor.

Heeltalle kan op die volgende twee maniere gespesifiseer word:

```
10 DEFINT A
20 FOR.....ens.

10    FOR A% = .....ens.
```

2.2. Die geneste lus

Dit is ook moontlik om verskeie lusse binne 'n lus te programmeer. So 'n stelsel van lusse word dan 'n geneste lus genoem. In die programuitvoering van 'n geneste lus word die binneste lus eers afgehandel voordat die program na die eersvolgende buitelus vorder, ens. totdat alle elemente van die geneste lus afgehandel is, waarna die res van die program dan normaalweg uitgevoer word. 'n Voorbeeld van 'n geneste lus volg hieronder:

```
10 FOR A = 1 TO 5
20 FOR B = 5 TO 1 STEP -1
30 PRINT B,A
40 NEXT: NEXT
```

Die syfers vanaf 5 tot 1 verskyn 5 keer links of die monitor, terwyl vyf 1'e, vyf 2'e, ens. regs verskyn.

Elke aparte lus moet natuurlik met 'n eie veranderlike geidentifiseer word. Die aantal lusse waaruit so 'n geneste lus bestaan word net deur die grootte van die rekenaar se geheue beperk.

Verskeie variasies met verskillende resultate is hier moontlik. Die moontlikhede hiervan is dus bykans onbeperk.

2.3. Die vertraaglus.

Hoewel uitvoering van 'n program baie vinnig plaasvind, neem dit tog werklike tyd in beslag. Weens die spoed waarmee stappe uitgevoer word, bestaan die behoefte om in sekere toepassings die uitvoering van die program vir 'n voorafbepaalde tyd te vertraag, bv. om besonderhede op die monitor vir 'n sekere tyd te behou voordat dit uitgewis moet word om vir ander besonderhede plek te maak. Die vertraaglus is 'n baie bruikbare metode om hierdie effek te verkry, soos in die volgende voorbeeld sal blyk:

```
10 PRINT "   Hierdie is ' vertraaglus"
20 FOR I = 1 TO 500 : NEXT
30 CLS
40 PRINT "   Sien jy hoe dit werk?"
50 FOR K = 1 TO 100 : NEXT
60 CLS
```

U sal merk dat die woorde in lyn 10 'n ruk lank op die monitor bly en verdwyn, waarna die woorde in lyn 40 verskyn en na 'n vyfde van die tyd verdwyn.

2.4. SLOT

Soos uit die voorgaande duidelik blyk, is die lus 'n uiters bruikbare tegniek met 'n wye toepassingsveld en waarsonder bykans geen program kan klaarkom nie.

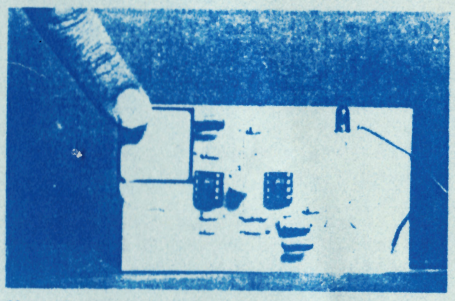
Touch Switch

For turning on equipment sensitive to vibration — photographic enlargers for example — or even just for a doorbell, this circuit is a boon

IF YOU'VE EVER got involved in repairing a transistor radio, you will probably know the trick of dabbing your finger at the input to the audio stage to see if you can hear the hum and/or increase in noise level to prove that the audio stage is working.

It is also possible to make use of this effect to switch a circuit positively. Switching by touch has become quite common in the last two or three years since the IC manufacturers have introduced devices specifically to perform this function. TV sets and lifts are amongst the common users (though some TV sets make use of your finger bridging two terminals and operate differently).

Touch switches may seem highly extravagant but industrially they are far more reliable, than mechanical types. In TV sets, touch switches have the great advantage of enabling the preset pots which control the varicap diodes in the timer to be put at the back of the set.



Above: the completed PCB. In this version we did not use a relay connecting a resistor and the LED in series across the output.

Parts List

RESISTORS. All ¼W. 5%

R1	56k
R2	100k
R3	10M
R4	15k
R5	15k
R6	3k3
R7	100k

CAPACITORS

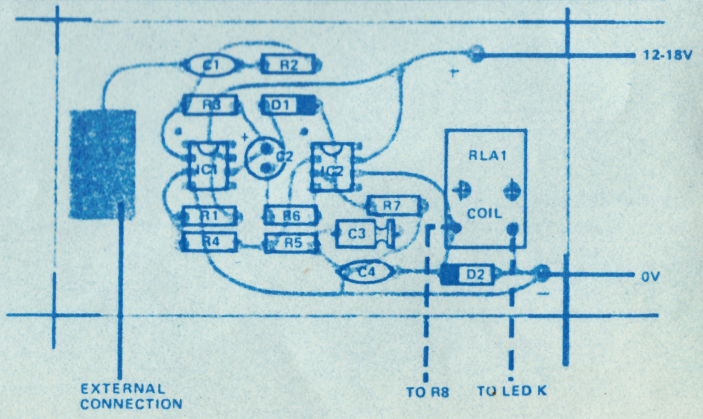
C1	100n polyester
C2	100uF 25 V electrolytic
C3	2u2 25 V electrolytic
C4	100n ployester

SEMICONDUCTORS

IC1, IC2	741 Op Amp
D1, D2	1N4148 silicon diode

MISCELLANEOUS

PCB as pattern, Miniature relay (500 ohm or more coil and change-over contacts).



On the right is the component overlay for the PCB. Be sure to position the polarised components C2, C3, IC1, IC2, D1 and D2 the right way round. If you're buying a relay get one to fit the holes as shown otherwise extra drilling and soldering may be needed.

How it Works

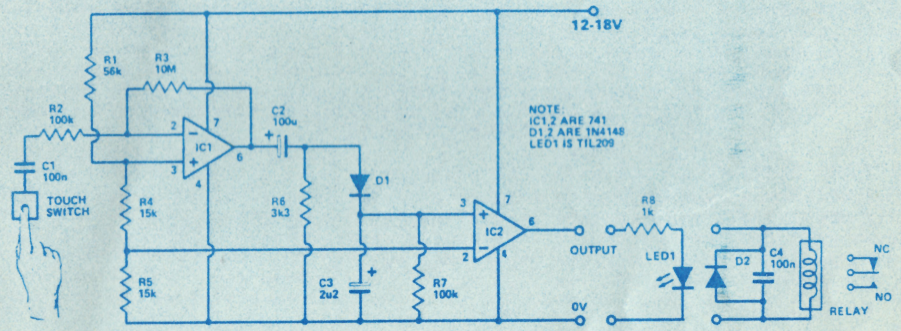
When a finger is placed on the touch contact, a low level of mains hum is introduced to IC1 via the DC blocking capacitor C1 and R2. The gain of IC2 is arranged at a very high level and the hum is thus enormous amplified and switches the output of IC1 to about 7 V peak-to-peak. The signal passes through C2 which blocks the DC to diode D1.

When the signal is swung positive, D1 conducts and charges up C3. When the voltage at the non-inverting terminal of IC2 (pin 3) exceeds the voltage at the inverting terminal(pin 2) the output of IC2 goes high and switches the relay.

IC2 acts as a voltage comparator. When the finger is removed, R7 discharges C3 and the output of IC2 goes low and the relay switches off.

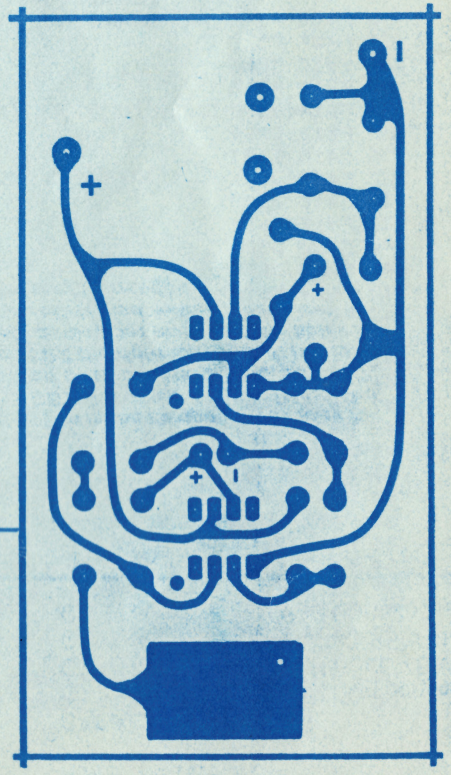
D3 and C4 are included to protect IC2 from the very high voltages caused by abruptly turning RL1 off.

If used to switch mains equipment, great care should be taken to isolate the switched mains from the rest of the circuit. R2 and C1 give only a margin of safety and are no substitute for taking care. The power for the circuit itself should only be derived from batteries.



Above is the circuit diagram, showing, on the right, the connections for either an LED (as in the photo on page 17), or a relay.

On the right is the copper foil pattern for the PCB we used, printed actual size.



CONSTRUCTION

We have designed our PCB so that the touch contract is on the board itself but have also allowed a terminal for taking the touch contact to somewhere more remote.

Small relays don't normally come in standard types and for that reason we have only left a general area showing the coil contacts, though these may have to be changed. We have not even attempted to show the switch contacts but there is plenty of room to run the contracts to terminals sited between the supply connections.

Apart from the obvious precautions to ensure that the ICs, diodes and the two electrolytic capacitors (C2, C3) are connected the right way around, construction is easy.

For safety reasons, we strongly recommend that the circuit is battery operated. If it is used to switch a relay, the contacts of the relay can be used to switch a mains supply on.

WHENEVER you look at a piece of work and you think the fellow was crazy, then you want to pay some attention to that. One of you is likely to be, and you had better find out which it is. It makes a lot of difference. —Charles Kettering

```
                ;*******************************
                ;* A SOFTWARE RESTART KEY FOR  *
                ;*          SVI 328            *
                ;* BY CHRIS VAN [illegible]OTHUIZEN*
                ;*******************************

START:          EQU   0F400H

HKEYI:          EQU   0FE79H
READY:          EQU   09AFH
ENDSTR:         EQU   0F546H
CTRLY:          EQU   6
CTRLZ:          EQU   2
SELECTY:        EQU   8
SELECTZ:        EQU   16

                ORG   START

;DIVERT INTERRUPT ROUTINE
;TO SCAN
F400 210EF4             LD    HL,SCAN
F403 F3                 DI
F404 227AFE             LD    (HKEYI+1),HL      ;ADDR
F407 3EC3               LD    A,0C3H            ;JP INSTR
F409 3279FE             LD    (HKEYI),A
F40C FB                 EI
F40D C9                 RET

;CHECK FOR CTRL-SELECT
SCAN:           DI
F40E F3         PUSH AF
F40F F5         LD    A,CTRLY
F410 3E06       OR    16
F412 F610       OUT   (96H),A
F414 D396       IN    A,(99H)
F416 DB99       AND   CTRLZ
F418 E602       JR    NZ,NOTRST
F41A 2021       LD    A,SELECTY
F41C 3E08       OR    16
F41E F610       OUT   (96H),A
F420 D396       IN    A,(99H)
F422 DB99       AND   SELECTZ
F424 E610       JR    NZ,NOTRST
F426 2015

;INCREMENT COUNT-IF COUNT IS 5
;THEN GOTO OK MESSAGE ELSE RET
F428 3A45F4             LD    A,(COUNT)
F42B 3C                 INC   A
F42C 3245F4             LD    (COUNT),A
F42F FE05               CP    5
F431 2010               JR    NZ,NOT5
F433 2A46F5             LD    HL,(ENDSTR)
F436 2B                 DEC   HL
F437 2B                 DEC   HL
F438 F9                 LD    SP,HL
F439 FB                 EI
F43A C3AF09             JP    READY

;CTRL-SELECT NOT PRESSED SO
;ZERO COUNT AND RETURN
F43D AF         NOTRST: XOR   A
F43E 3245F4             LD    (COUNT),A
F441 F1                 POP   AF
F442 C9                 RET

                ;RETURN LEAVING COUNT AS IS
F443 F1         NOT5:   POP   AF
F444 C9                 RET

F445 00         COUNT:  DB    0

                        END
```

HOW TO USE THE PROGRAM

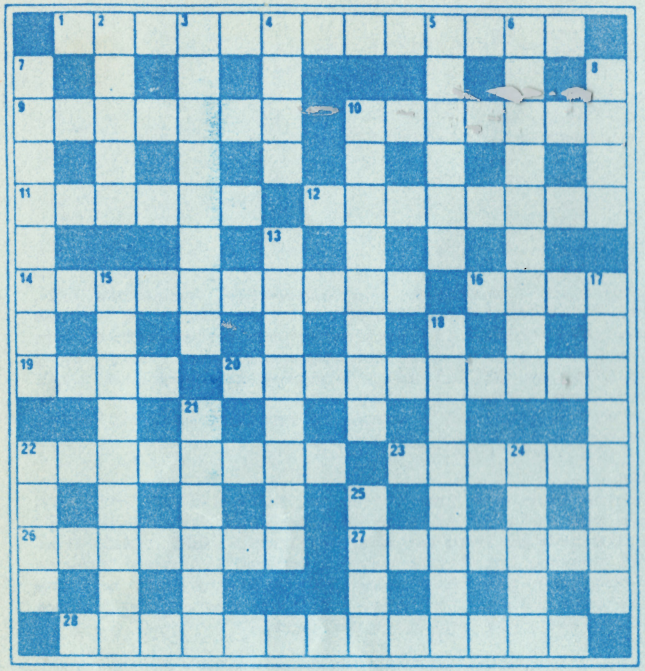
Type in the program changing syntax wherever your assembler requires it. Assemble and BSAVE"rstkey",&HF400,&HF445,&HF400. To use it type CLEAR200,&HF3FF:BLOAD"rstkey",R(ENTER). If you don't have an assembler you can poke the hex code in the listing into the addresses as specified, then BSAVE as above. Pressing CTRL and SELECT after running the program will now return you to the OK message, provided that interrupts have not been disabled. Note that this is different from a hardware restart switch, which reboots the computer, so maybe this is even more handy.

POSSIBLE USE OF THE PROGRAM

When programming in machine code and you think you have fallen into an endless loop, this will save you lots of time otherwise used for reloading. BEWARE - If your program did not go into an endless loop, but has lost control completely, several memory locations may be messed up. Your source code may still look OK but your assembler may be messed up, or the program/routine that writes your source to tape. So remember, don't save your new source file over the last copy you have on tape!

In an apple-eating contest Sue ate ninety apples and Selena ate a hundred and one. How many more apples did Selena eat than Sue?

Answer: Ten — Selena ate a hundred and *won!*



ACROSS

1. It's a language-expanding process (4-9)
9. Drunkard surpasses drunkard, in retrospect (7)
10. Distressing description of well-stocked French bakery? (7)
11. Regular patronage is the usual practice (6)
12. He could make Len groan (8)
14. Where the roof is paid for by the publican (2, 3, 5)
16. Superficial form of entertainment (4)
19. Translate 'go sour' (4)
20. Two kings, one verse writer (6, 4)
22. Where the members provide shortened clue to 'sweeper' (8)
23. Saleroom muddle? (6)
26. The first coats shown in the text-books (7)
27. A scandalous thing it's not in the gearshift (7)
28. Event of 1926 leaving the army leaderless? (7, 6)

DOWN

2. Growth area in deserted surroundings (5)
3. Disapproved strongly of department leaders getting older letters (8)
4. Those felt to be frisky? (4)
5. Wound round the newt I'd dissected (6)
6. The attack is not at all pleasant (9)
7. Are very obvious when they protrude (5, 3)
8. A little eagle nesting in a Scottish valley (4)
10. Meeting of papal compatriots and sailor is of astronomical significance (4, 4)
13. To mistakenly put ours down is amazing (8)
15. Turns out risqué and colourful (9)
17. Newly sold this way in Suffolk? (8)
18. Man associated with an insurance company loss? (8)
21. Country makes great start with EEC regrouping (6)
22. Imitate the twenty-fifth policeman (4)
24. You won't be successful in drawing one (5)
25. No applause for the liquor is heard (4)

****
ANSWERS ELSEWHERE
IN THIS NEWSLETTER
****

Our youngest son, just turned 16, proudly applied for his first real job. He was puzzled by some of the questions asked in the interview, but had no difficulty in answering the final one: "What is the one thing about yourself you would change if you could?"

Charles's answer: "My bank balance."

He started work the next day.

—Shirley Allen

Toevoereenhede

Die belangrike dele van die mikrorekenaar – die sentrale verwerker en die hoofgeheue – is klein. Hulle pas op 'n klein gedrukte stroombaanbord. Die grootste gedeelte van die mikrorekenaar is die skerm en die toetsbord, die toevoer- en afvoereenhede. Toetsborde moet byvoorbeeld groot genoeg gebou word vir die mens se hande en vingers.

Die toetsbord

Die toetsbord lyk gewoonlik net soos 'n tikmasjien s'n; dus sal iemand wat tikmasjiene ken, maklik weet waar die letters is. Mikrorekenaars se toetsborde is egter ingewikkelder. Die rekenaar moet byvoorbeeld kan onderskei wanneer 'n gewone woord soos "druk" in 'n sin gebruik word (bv.: "Hierdie druk is te klein") en wanneer "druk" as 'n opdrag bedoel word.

Die normale tikmasjienletters en -syfers word dus aangevul met spesiale toetse om behulpsaam te wees met instruksies aan die rekenaar. Daar bestaan geen rede waarom toetsborde nie 'n toets kan hê vir alle opdragte wat 'n mens wil gee nie, maar sulke toetsborde sal te groot en lomp wees. In plaas daarvan word 'n kombinasie van toetse gebruik.

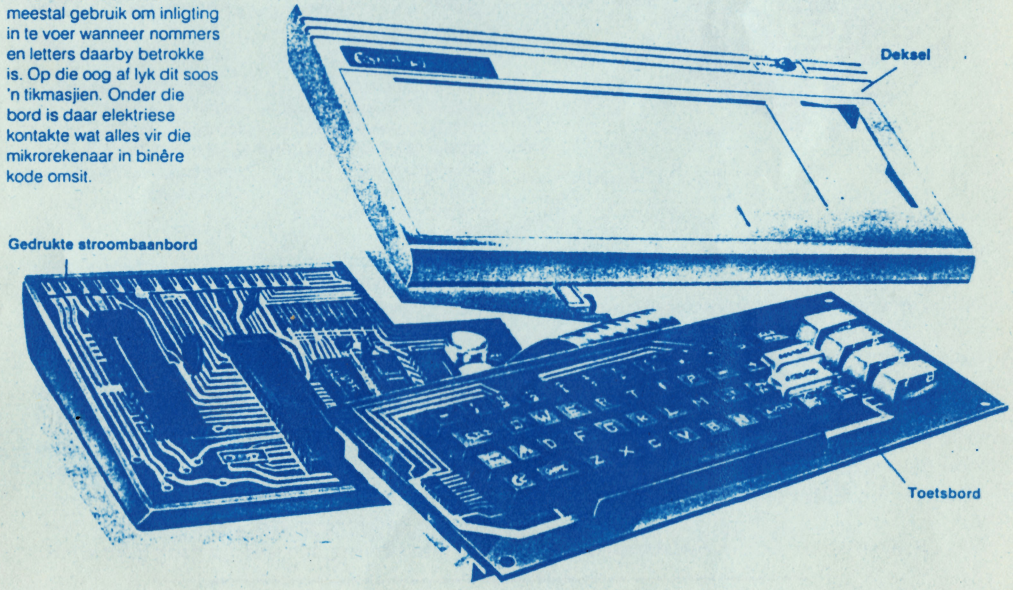
Van letters na binêre bisse

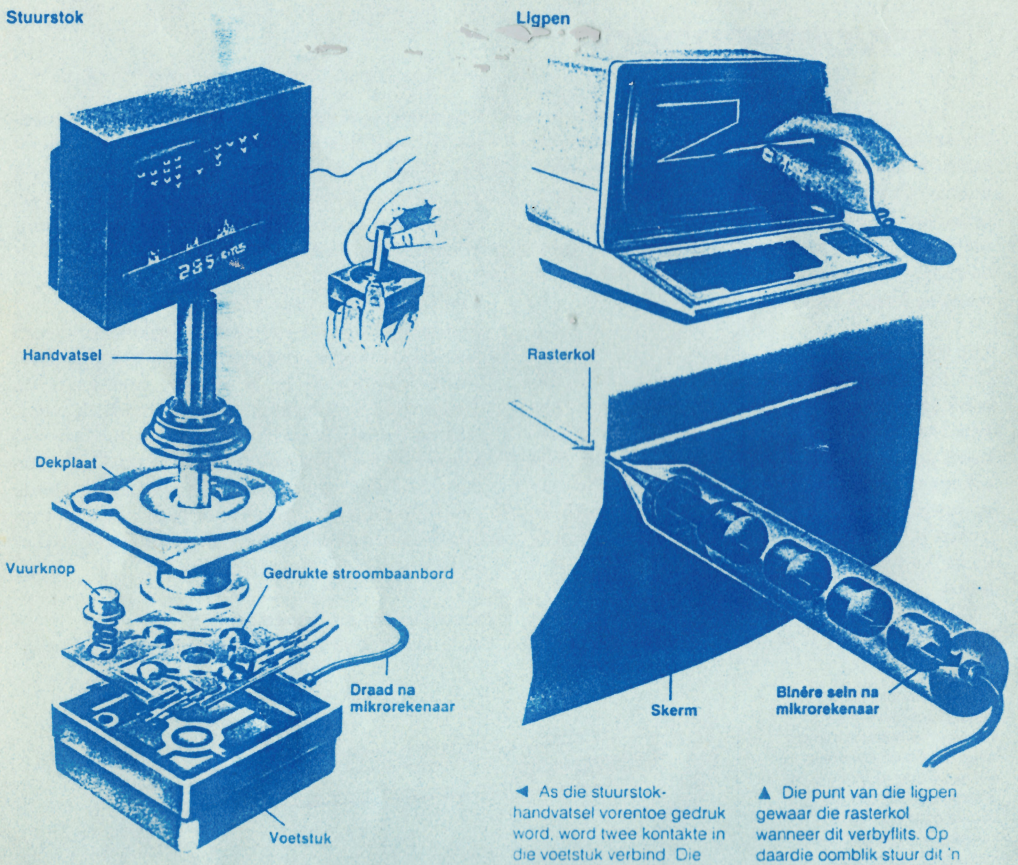
Wat ook al op die toetsbord getik word, word deur die mikrorekenaar in binêre bisse vertolk. Onder die toetse is 'n rooster met kruis en dwars drade. Wanneer 'n toets afgedruk word, raak 'n metaalkontak aan 'n kruispunt op die rooster. Die mikrorekenaar bepaal hierdie punt. Dit weet dat dit nou 'n spesifieke biskode moet "verstaan". Gewone letters, nommers en bevele word onmiddellik omgesit in die taal of kode wat die rekenaar verkies.

Stuurstokke en ligpenne

'n Toetsbord is slegs een manier waarop inlig-

► Die toetsbord word meestal gebruik om inligting in te voer wanneer nommers en letters daarby betrokke is. Op die oog af lyk dit soos 'n tikmasjien. Onder die bord is daar elektriese kontakte wat alles vir die mikrorekenaar in binêre kode omsit.





◄ As die stuurstok-handvatsel vorentoe gedruk word, word twee kontakte in die voetstuk verbind. Die mikrorekenaar moet geprogrammeer word om dit as 'n vorentoebeweging te vertolk.

▲ Die punt van die ligpen gewaar die rasterkol wanneer dit verbyflits. Op daardie oomblik stuur dit 'n binêre boodskap na die mikrorekenaar.

ting in 'n mikrorekenaar gevoer word. Wanneer *Space Invaders* gespeel word, is dit baie makliker om jou tuig te maneuvreer met iets wat die beweging op die skerm reflekteer. In hierdie geval is die stuurstok nuttig. Terwyl 'n mens die arm in die sok rondbeweeg, word die bewegings omgesit in binêre opdragte wat byvoorbeeld op, links, af, ens. beveel. Wanneer gevuur moet word, veroorsaak die knop wat 'n mens druk, die stippelpatroon vir die ontploffing.

Nog 'n alternatief vir die toetsbord is die ligpen. 'n Mens kan dit gebruik om vrae te beantwoord of om lyne oor die skerm te trek. Die ligpen funksioneer omdat 'n helder ligkol agter die skerm die pen se ligsensitiewe punt ontsteek. Die kol, bekend as die rasterkol, beweeg op 'n ordelike manier oor die agterkant van die skerm. As gevolg hiervan kan die mikrorekenaar uitreken waar die kol was toe die punt van die pen ontsteek het. Afhangende van hoe die mikrorekenaar geprogrammeer was, kan 'n handeling dan aan die gang gesit word.

AT A dinner party we should eat wisely, but not too well, and talk well, but not too wisely.

–Somerset Maugham

MSX MSX MSX MSX
ONLY ONLY ONLY

NOW YOU CAN SAVE AND LOAD YOUR PICTURES.. TO & FROM CASSETTE !

TO SAVE THE SCREEN TO TAPE.....

```
10 COLOR15,0,0:KEYOFF:CLS:MOTOROFF
20 LOCATE0,10,0:PRINT"PRESS PLAY AND RECORD ON TAPE"
30 PRINT"       and ENTER"
40 IFINKEY$=""THEN40
50 COLOR15,0,0:SCREEN2
60 DEFINTA-Z
70 ' ENTER LINES FOR PICTURE FROM HERE
80 '
90 '
100 '
110 '
120 FORA=&H0T0&H3800:Z=VPEEK(A):POKEA+&HA000,Z:NEXTA
130 BSAVE"CAS:PICT",&HA000,&HD800
140 GOTO140
```

TO LOAD THE SCREEN FROM TAPE.....

```
10 CLEAR200,&HA000
20 COLOR15,0,0:KEYOFF:CLS:MOTOROFF
30 LOCATE0,10,0:PRINT"PRESS PLAY ON TAPE"
40 PRINT"       and ENTER"
50 IFINKEY$=""THEN50
60 COLOR15,0,0:SCREEN2
70 DEFINTA-Z
80 BLOAD"CAS:PICT"
90 FORA=&H0T0&H3800:Z=PEEK(A+&HA000):VPOKEA,Z:NEXTA
100 GOTO100
```

I saw Esau sitting on a see-saw — how many S's in that?
There aren't any S's in THAT!

HANDY EXTRAS

If you're not too hot at typing, don't despair – there's plenty of alternatives to lend a helping hand

You don't need to be an expert typist to be able to use your MSX micro. By making use of the joystick ports and plugging in such devices as a mouse or tracker ball you can by-pass the keyboard altogether.

So, if you have an aversion to using the keyboard, or are a hopeless typist don't panic — we've rounded up a selection of input devices that are easy to use and can make home computing simple and fun.

You're probably already familiar with joysticks as games controllers, but did you know that they can also be used for manipulating text and graphics?

With graphics software packages such as Electric Studio's *Eddy II* and Kuma's *Colour Fantasia*, you can forget about fiddling around with the cursor control keys and have more direct control simply by moving a stick around.

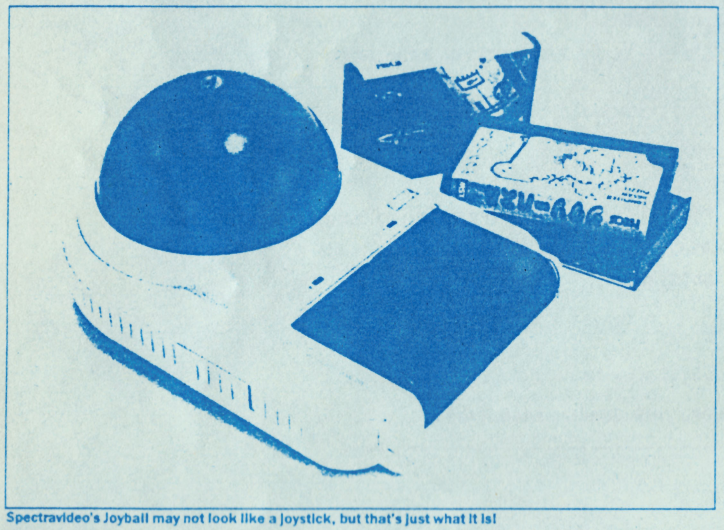
Word processing can be made so much more simple too. Although you still have to input text via the 'Qwerty' keyboard, instead of fiddling with the cursor controls to move paragraphs or alter the spacing between words you just move your joystick, press one of the fire buttons and away you go.

Joysticks can come in all shapes and sizes and some, like Suncom's Joy Sensor don't even have a 'stick' as such but a touch sensitive pad.

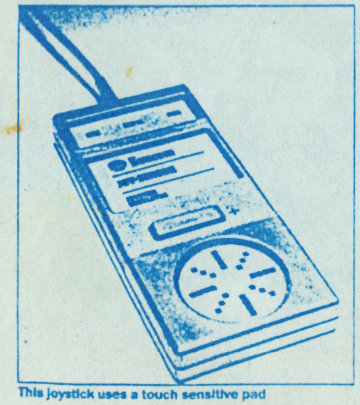
Others, like Spectravideo's Joyball have a rotating ball mounted on a square plastic casing. The beauty of this kind of device is that it offers you multi-directional movement, whereas most joysticks limit you to the eight compass points instead of 16 (which can be very frustrating).

Another type of joystick that's easy to use is Sony's JS-75 infra-red joystick controller. Although it is a bit pricey it means that you have complete freedom of movement, since no lead is required between the joystick and computer. Obviously it's great for playing games but it is also equally at home when used together with a word processing package or for moving sprites across the screen when designing graphics.

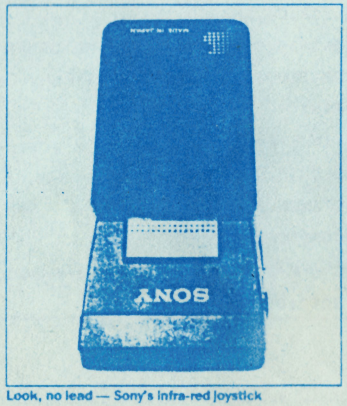
A more sophisticated and specialised device is a tracker ball. Marconi markets one for MSX micros. It is known as the RB2



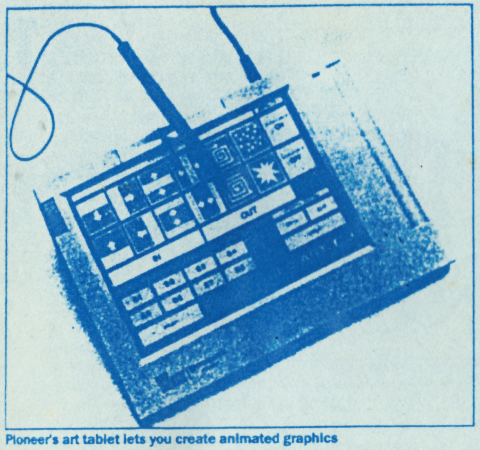
Spectravideo's Joyball may not look like a joystick, but that's just what it is!



This joystick uses a touch sensitive pad



Look, no lead — Sony's infra-red joystick



Pioneer's art tablet lets you create animated graphics

It is a simple to use device that can be used for games playing as well as creating computer graphics.

All you have to do is plug the tracker ball lead into the joystick port of your computer. In a few seconds the menu appears and then, by gently rolling the ball with your fingertips, you can move the cursor over the menu and pick out which colour you want to draw in and the function you wish to use. It is as simple as that.

The software provided with the tracker ball gives you a palette of 15 colours from which to choose. You can create just about any shape you like — circles, squares, rectangles, triangles and straight lines are all possible.

By using the text mode you can use the keyboard to create characters, and the tracker ball to place them anywhere on the screen — particularly useful for technical artists who need to label their drawings.

You can magnify specific portions of a graphic design and make modifications pixel by pixel, and if you're not satisfied with the end result the 'eraser' facility will rub it all out.

The tracker ball isn't the only device with these kinds of capabilities. Wigmore House has brought out a very compact little gadget called the MS2000 mouse, a cassette-based graphics package called *Cheese*.

A mouse is basically an upside down version of the tracker ball, with the ball placed underneath the casing. It is operated by simply pushing the mouse around on a flat surface. A cursor, or pointer, moves in the corresponding direction on screen.

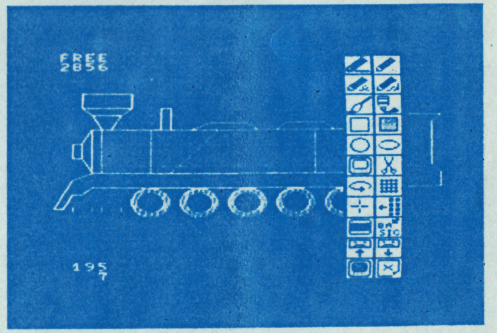
Cheese's menu is much larger than that of the tracker ball package. For starters you get not only the 15 primary colours, but an additional 120 different shades. Graphics can be scrolled up and down the screen as well as backwards and forwards.

All the usual shape commands are included — squares, circles, ellipses, rectangles and triangles, etc. And you can save all your creations to floppy disk or cassette tape.

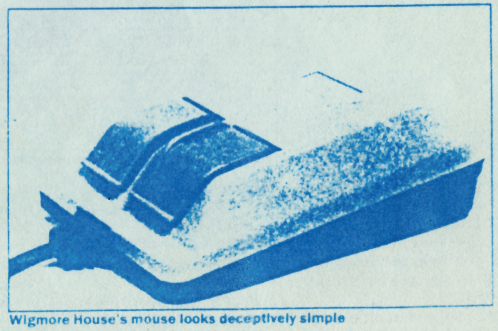
We've not heard of any MSX owners using the mouse for

The high quality graphics possible with a tracker ball

A first effort with Sanyo's light pen



The Marconi tracker ball menu icons



Wigmore House's mouse looks deceptively simple

word processing, as yet. However, Microsoft, creator of the MSX standard, sells a word processing package called *Word* bundled together with a mouse for the IBM PC, so maybe it won't be too long before the appropriate software appears for MSX users.

Another device which bypasses the keyboard is the lightpen from Sanyo. It comes complete with its own software. Unlike the two devices we've looked at so far, the lightpen can only be used for the input of graphics — not for text. A lightpen is very simple to use. The software is included in the cartridge interface. Just plug it in and wait for the menu to appear.

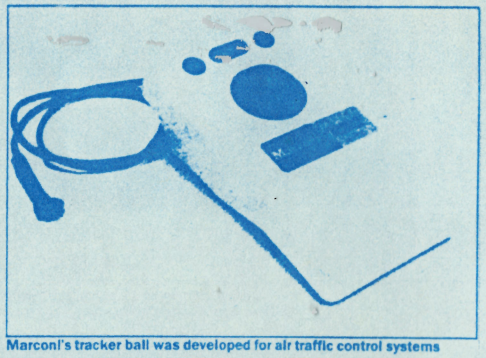
Using the light pen is just like drawing with a large wax crayon. You create shapes and drawings simply by moving the tip of the pen around your TV screen. Like the tracker ball and mouse it has all the drawing features you'd expect of a graphics tool. There is one problem, though, which is inherent in its design.

After continued use you might develop some cramp in your hand and arm and you will probably find your arm partly obscures your field of vision.

Zoom

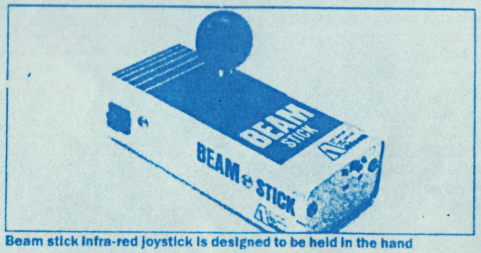
Having said that, the advantage of this device is that its software allows you to convert your graphics creation into BASIC and run it off as a listing. Like the tracker ball, there's an enlargement facility whereby you can zoom in on a particular section of your drawing and work on it pixel by pixel.

A similar device to Sanyo's lightpen is the Grafpad from British Micro.



Marconi's tracker ball was developed for air traffic control systems

Toshiba's MSX joystick has rubber feet, two fire buttons



Beam stick infra-red joystick is designed to be held in the hand

In terms of design, the Grafpad is probably the most 'natural' peripheral for artists to use. It consists of a flat electronic pad covered in graph lines with an electronically sensitive pen attached. Using it is literally like putting pen to paper.

You simply draw on the pad with the pen and the design appears on the screen.

It is more versatile in use than a lightpen because you are drawing on a flat surface, without having to reach up to the screen.

The only obvious drawbacks are that it cannot draw 'real' circles — we had to make do with ellipses, and you have to fill in a shape with the same colour as its border, or the colour 'bleeds' all over the screen. The problem with circles is probably due to the MSX computer, not the Grafpad.

At the time of writing, British Micro's Frank Sketch tells us that the software for the Grafpad is being re-written and that the new version will have as many features as the lightpen.

Any device that bypasses a keyboard is obviously going to be useful not only for young children (particularly the handicapped) but for anyone who cannot type.

Joysticks are cheap and underrated — they could easily be used for manipulating chunks of text in word processing or with spreadsheets. However, they're not used in this way because business users tend to associate them solely with games playing.

This doesn't apply to the tracker ball. Marconi originally developed it for the specialised field of cursor movement control on air traffic control displays.

Verdict

Manufacturers are constantly looking out for alternative methods of inputting information into micros, in a never-ending bid to make computing more 'user-friendly' for the end user.

In the non-MSX business world there are such devices, such as the Pen Pad which is similar to the Grafpad and is solely for word processing.

All the devices we've looked at are mainly graphics-orientated, but the tracker ball and the mouse can be used for playing games too.

The decision as to what peripheral you might need will depend on what you are going to use it for and how much you are prepared to pay.

If you are just looking for something to zap aliens with instead of juggling around with the cursor control keys then a joystick is ideal and within most people's budgets.

But if you are interested in computer graphics then you have the mouse, tracker ball, light pen and Grafpad devices to choose from.

Whichever device you decide upon, try it out at your dealer before handing over the money. Better still, take your computer with you to make absolutely sure you will know what to do with the device once you get home. ■

```
***   MSX     MSX     MSX   ***
        ONLY    ONLY    ONLY
**************************************
```

```
10 PRINT"********************************
*******"
20 PRINT"**         M.S.X  FROGGER
   **"
30 PRINT"**        BY CLIFF WALTERS
   **"
40 PRINT"**         MARCH-MAY  1985
   **"
50 PRINT"**        ON SPECTRAVIDEO.
   **"
60 PRINT"**             SVI  728
   **"
70 PRINT"********************************
*******"
80 CLEAR500,&HEA2E
90 ONSTOPGOSUB2130:STOP ON
100 CLS:KEYOFF:GOSUB 2050
110 N=RND(-TIME):GOSUB 1500
120 SCREEN1,2:COLOR 14,1,1:WIDTH 32
130 DEFUSR0=&H41
140 DEFUSR1=&H44
150 DEFUSR2=&HEA2F
160 DEFFNA(X)=INT(RND(1)*X)+1
170 GOSUB 830:GOSUB 1700:HI=0
180 LE=1:SC=0:Z=100
190 FL=0:FH=0:LI=5
200 CLS:PRINT:PRINT:PRINT"FROGGER LEVEL
":PRINT
210 FORI=4TO19:PRINTSPC(I)LE:NEXT I
220 LOCATE 0,21:PRINT"SCORE=";SC
230 GOSUB 1460 :GOSUB 820 :GOSUB 820
240 U=USR0(0):CLS:LOCATE 0,0 :PRINT"SCOR
E ";SC:LOCATE13,0:PRINT"HI=";HI :LOCATE
24,0 :PRINT"LIVES";LI
250 LOCATE 0,1:PRINT STRING$(32,216)::LO
CATE 0,2:PRINT"bbb`abbbb`abbbb`abbbb`abb
bb`abbb";
260 LOCATE0,3:PRINTSTRING$(32,145)
270 FOR I=3 TO 27 STEP 6:LOCATE I,3:PRIN
TSPC(2):NEXT I
280 LOCATE0,4:PRINT  STRING$(128,147)
290 LOCATE0,8:PRINT  STRING$(32,144)
300 LOCATE0,9:PRINT  STRING$(32,98)
310 LOCATE0,10:PRINT  STRING$(32,131)
320 LOCATE0,11:PRINT  STRING$(32,128)
330 LOCATE0,12:PRINT  STRING$(32,130)
340 LOCATE2,14:PRINT I$
350 LOCATE0,16:PRINT  STRING$(32,129)
360 LOCATE0,17:PRINT  STRING$(32,130)
370 LOCATE0,19:PRINT I$
380 LOCATE0,21:PRINT STRING$(32,129)
390 FOR I=6848 TO 6879 :VPOKE I,136:NEXT
 I
400 FOR I=6880 TO 6911 :VPOKE I,98:NEXTI
:SP=0:M=0
410 X=16:Y=22:PUTSPRITE 1,(X*8,Y*8),15,4
420 ON SPRITE GOSUB 810:SPRITE ON
```

```
430 IF M=1 THEN 490
440 LOCATE0,4:PRINTA$:LOCATE0,5:PRINT B$
450 LOCATE0,7:PRINTC$:LOCATE0,6:PRINT D$
460 LOCATE0,13:PRINTE$:LOCATE0,15:PRINTF
$
470 LOCATE0,18:PRINTG$:LOCATE0,20:PRINTH
$
480 U=USR1(0)
490 FOR D=1 TO Z:NEXT D
500 U=USR2(0)
510 PUTSPRITE0,(SP,66),15,SP MOD2+1
520 SP=SP+5:S=STICK(0)
530 X=X+(S=7 AND X>0)-(S=3 AND X<31)
540 IF S=1 THEN Y=Y-1:PUTSPRITE 1,(X*8,Y
*8),,7:SC=SC+10:LOCATE 5,0:PRINTSC
550 PUTSPRITE 1,(X*8,Y*8),,4-(S=3)+(S=7)
560 IF VPEEK(Y*32+X+6144)>146 THEN 810
570 IF Y>3 THEN 490
580 IF VPEEK(Y*32+X+6144)<>32 THEN810
590 IF X=4 OR X=10 OR X=16ORX=22 OR X=28
 THEN X=X-1
600 PUTSPRITE 1,(-16,0):LOCATE X,Y-1:PRI
NT"cd":LOCATE X,Y:PRINT"ef"
610 SC=SC+100:LOCATE 5,0:PRINTSC
620 PLAY "T255L6403C6404C6405C6406C64"
630 FH=FH+1:IF FH=5 THEN 720
640 M=1:GOTO 410
650 FOR I=1 TO 5:PUTSPRITE1,,FNA(15),3
660 FOR DL=1 TO 30:NEXT DL
670 PUTSPRITE1,,FNA(15),6
680 FOR DL=1 TO 30:NEXT DL,I
690 PUTSPRITE 1,(-16,0):LI=LI-1
700 LOCATE 29,0:PRINT LI:IF LI=0 THEN750
710 M=1:GOTO 410
720 GOSUB 820:SC=SC+1000:LE=LE+1:PUTSPRI
TE 0,(-16,0)
730 Z=Z-10:IFZ<=0THENZ=0
740 GOTO 190
750 IFSC>HITHENHI=SC:LOCATE16,0:PRINTHI:
FL=1
760 GOSUB820:PUTSPRITE0,(-16,0):CLS:PRIN
T:PRINT:PRINT SPC(6)"YOUR SCORE WAS";SC:
IFFL=1THEN780
770 IFFL=0THENLOCATE8,7:PRINT"HISCORE=";
HI:GOTO 790
780 PRINT:PRINT:PRINT:PRINT SPC(4)"A NEW
 HI SCORE OF";HI:FL=0
790 LOCATE1,20:PRINT"PRESS SPACE BAR TO
PLAY AGAIN"
800 IF NOT STRIG(0)THEN800ELSE180
810 SPRITE OFF:PLAY "T255L6406C6405C6404
C6403C6402C64":GOTO 650
820 FOR DL=1TO 750:NEXT DL:RETURN
830 LOCATE 8,20:PRINT"ONE MOMENT PLEASE"
840 RESTORE 880
850 FOR I=1 TO 7:SP$="":READ S$:FOR B=1
TO LEN(S$) STEP 2
860 SP$=SP$+CHR$(VAL("&H"+MID$(S$,B,2)))
:NEXT B
870 SPRITE$(I)=SP$:NEXT I
880 DATA 00000C3FF3C0000000000000000000
00000C0B0E0000000
890 DATA 00000C3FC3000000000000000000000
000000C3BFEC00000
900 DATAC2251C7E7E1C25C2
910 DATA81995A3C3C7E9942
920 DATA43A4387E7E38A443
930 DATAC3E77E24247EE7C3
940 DATA004281995A3C3C3C1824244242240000
950 C=0:READ A:IF A=999 THEN RETURN
960 READ C$:FOR B=1 TO LEN(C$) STEP 2
970 VPOKE A*8+C,VAL("&H"+MID$(C$,B,2)):C
=C+1:NEXT B
980 GOTO 950
990 DATA 33,0000009900000000
1000 DATA 40,3C7EFFE7E7FF7E3C
1010 DATA 96,FFF0C08080000000
1020 DATA 97,FF0F030101000000
1030 DATA 98,FFFFFFFFFFFFFFFF
1040 DATA 99,FFF1D993950B0F07
1050 DATA 100,FF0F9BC9A9D0F0E0
1060 DATA 101,0703070D0C384400
1070 DATA 102,E0C0E0B0301C2200
1080 DATA 104,3F535C8B8C535C3F
1090 DATA 105,FF33CC33CC33CCFF
1100 DATA 106,FC32CE33CD32CEFC
1110 DATA 112,3F535C8B8C535C3F
1120 DATA 113,FF33CC33CC33CCFF
1130 DATA 114,FC32CE33CD32CEFC
1140 DATA 120,187E7EFFFF7E7E18
1150 DATA 128,8181FF8181FF81FF
1160 DATA 129,FFFF000000000000
1170 DATA 130,000000000000FFFF
1180 DATA 131,0000000000000000
1190 DATA 136,000044D26EEEFFFF
1200 DATA 144,000044D26EEEFFFF
1210 DATA 145,FFFDBE2934440000
1220 DATA 146,815A3C7E7E3C5A81
1230 DATA 147,0000000000000000
1240 DATA 184,FF80BFBFBFBF80FF
1250 DATA 192,077EF3F3F3F37E07
1260 DATA 193,F83EE7E7E7E73EF8
1270 DATA 194,E02EBDB9B9BD2EE0
1280 DATA 195,78337FCECE7F3378
1290 DATA 196,008EE43F3FE48E00
1300 DATA 197,0775BDBFBFBD7507
1310 DATA 200,E04779F9F97947E0
1320 DATA 201,7EFFF7F7F7F7FF7E
1330 DATA 208,077EF3F3F3F37E07
1340 DATA 209,F83EE7E7E7E73EF8
1350 DATA 210,E02EBDB9B9BD2EE0
1360 DATA 211,78337FCECE7F3378
1370 DATA 212,008EE43F3FE48E00
1380 DATA 213,0775BDBFBFBD7507
1390 DATA 216,DFDF00FBFBFB00DF
```

```
1400 DATA 217,FF80BFBFBFBF80FF
1410 DATA 218,00AAAA0000AAAA00
1420 DATA 227,FF5555FFFF5555FF
1430 DATA 233,FF5555FFFF5555FF
1440 DATA 252,00AAAA0000AAAA00
1450 DATA 999
1460 RESTORE 1490
1470 READ A:IF A=999 THEN RETURN
1480 READ B:VPOKE 8192+A,B:GOTO 1470
1490 DATA 5,142,12,192,13,78,14,30,15,22
2,16,26,17,42,18,206,23,78,24,96,25,208,
26,112,27,182,28,206,29,42,30,32,31,26,9
99
1500 REM TITLE
1510 SCREEN 3:COLOR 1,1,1:CLS
1520 OPEN "GRP:"AS#1
1530 FOR H=1 TO 10:J$="M.S.X":J=52:K=24:
FOR I=1 TO 2
1540 FOR L=1 TO LEN(J$):COLOR RND(1)*14+
2
1550 PRESET(J,K):PRINT#1,MID$(J$,L,1)
1560 J=J+32:NEXT L:J$="FROGGER"
1570 J=20:K=112:NEXT I,H
1580 GOSUB 820
1590 REM INSTRUCTIONS
1600 SCREEN 0:WIDTH 40:COLOR 14,1,1
1610 PRINTSPC(14)"INSTRUCTIONS",SPC(14)"
~~~~~~~~~~~~":PRINT
1620 PRINT" USING THE CURSOR KEYS, GUIDE
 THE FROG":PRINT"ACROSS THE BUSY MOTORWA
Y AND THE RIVER"
1630 PRINT"IN ORDER TO REACH YOUR ARCHWA
Y HOME,":PRINT:PRINT"YOU GET 10 POINTS F
OR EVERY STEP MADE"
1640 PRINT"AND 100 POINTS FOR EVERY FROG
 HOME PLUS":PRINT"A BONUS OF 1000 POINTS
 FOR 5 FROGS HOME ":PRINT"FROM WHERE, YO
U GO ON TO THE NEXT LEVEL.":PRINT
1650 PRINT"  EACH LEVEL GETS A LITTLE BI
T FASTER.":PRINT:PRINT"KEEP YOUR FROG MO
VING WHEN ON THE LOGS":PRINT"OR MULTI-CO
LOURED TURTLES ETC.":PRINT"OR IT WILL F
ALL IN AND DROWN.":PRINT
1660 PRINT"ALSO AVOID HITTING THE FAR RI
VER BANK ":PRINT"OR BEING EATEN BY THE S
LIPPERY SNAKE.":PRINT
1670 PRINT"PRESS THE SPACE BAR TO START
PLAYING"
1680 IF NOT STRIG(0)THEN 1680
1690 RETURN
1700 RESTORE 1900
1710 X$="":FORA=1TO32:READB
1720 X$=X$+CHR$(B):NEXTA:A$=X$
1730 X$="":FORA=1TO32:READB
1740 X$=X$+CHR$(B):NEXT A:B$=X$
1750 X$="":FORA=1TO32:READB
1760 X$=X$+CHR$(B):NEXTA:C$=X$
1770 X$="":FORA=1TO32:READB
1780 X$=X$+CHR$(B):NEXTA:D$=X$
1790 X$="":FORA=1TO32:READB
1800 X$=X$+CHR$(B):NEXTA:E$=X$
1810 X$="":FORA=1TO32:READ B
1820 X$=X$+CHR$(B):NEXTA:F$=X$
1830 X$="":FORA=1TO32:READ B
1840 X$=X$+CHR$(B):NEXTA:G$=X$
1850 X$="":FORA=1TO32:READ B
1860 X$=X$+CHR$(B):NEXTA:H$=X$
1870 I$="!   !   !   !   !   !   !"
1880 RETURN
```

```
*****************************************
*  I M P O R T A N T    N O T I C E   *
*  ---------------------------------   *

*  1) BEFORE TYPING A GAME PROGRAM,    *
*     ALWAYS FIRST TYPE AS A DIRECT    *
*     COMMAND.......                   *
*        COLOR15,0,0:WIDTH 40          *
*     THIS WILL GIVE A GOOD CONTRAST   *
*     AND THE PROGRAM WILL APPEAR      *
*     ON YOUR MONITOR EXACTLY AS IT    *
*     IS LISTED IN THE NEWSLETTER.     *
*  2) AS EACH OF THESE GAME PROGRAMS   *
*     IS FULLY TESTED, RUN AND THEN    *
*     PRINTED OUT ON A LINE-PRINTER,   *
*     THE CLUB DOES NOT WELCOME ANY    *
*     TELEPHONE CALLS, LETTERS OR      *
*     REQUESTS FOR TAPES.              *
*  3) IN VIEW OF THE POOR QUALITY      *
*     OF THE PRINTING OF 'FROGGER'     *
*     IN LAST MONTH'S NEWSLETTER, WE   *
*     HAVE DECIDED TO PRESENT IT IN    *
*     OUR 'STANDARD' FORMAT IN THIS    *
*     MONTH'S NEWSLETTER AGAIN.        *
*****************************************
```

A deaf and dumb man went into an ironmonger's shop to buy a hammer. He couldn't speak so he mimed hammering... then another deaf and dumb man went in wanting a screwdriver, so he mimed putting in a screw... Then a blind man went into the shop and he wanted a saw. So what sort of mime did he do?

He didn't mime at all — he just asked for it!

```
1890 REM A$
1900 DATA 112,113,113,113,114,147,147,14
7,147,147,147,147,104,105,105,106,147,147,14
7,147,147,147,104,105,105,105,106,147,14
7,147,147,147,147
1910 REM B$
1920 DATA 147,147,147,147,147,146,40,40,
146,147,147,147,147,147,147,147,40,120,4
0,146,147,147,147,147,147,147,147,120,40
,146,40,120
1930 REM C$
1940 DATA 40,146,120,120,146,40,147,147,
147,147,147,147,146,40,120,40,147,147,14
7,147,147,120,40,146,146,40,120,147,147,
147,147,147
1950 REM D$
1960 DATA 147,147,147,147,147,104,105,10
5,105,105,106,147,147,147,147,112,113,11
3,113,114,147,147,147,147,112,113,113,11
3,113,113,113,114
1970 REM E$
1980 DATA 32,32,32,208,209,32,32,32,197,
233,233,233,32,32,32,32,197,252,252,32,3
2,32,32,200,201,32,32,32,213,217,217,217
1990 REM F$
2000 DATA 208,209,32,32,32,32,218,218,21
8,218,210,32,32,32,32,32,32,32,195,196,3
2,32,32,32,184,184,184,194,32,32,32,32
2010 REM G$
2020 DATA 197,184,184,184,184,32,32,32,3
2,32,200,201,32,32,32,32,213,227,227,227
,227,32,32,32,32,32,192,193,32,32,32,32
2030 REM H$
2040 DATA 32,32,32,211,212,32,32,32,32,2
33,233,233,194,32,32,32,32,32,192,193,32
,32,32,32,32,218,218,218,218,210,32,32
2050 RESTORE 2100
2060 FOR A%=&HEA2FTO&HEAB8
2070 READ B$
2080 POKEA%,VAL("&H"+B$)
2090 NEXT A%:RETURN
2100 DATA 3E,05,32,50,C3,3E,04,32,51,C3,
CD,60,EA,3E,07,32,50,C3,3E,06,32,51,C3,C
D,60,EA,3E,0F,32,50,C3,3E,0D,32,51,C3,CD
,60,EA,3E,14,32,50,C3,3E,12,32,51,C3
2110 DATA 3A,50,C3,01,20,00,21,1F,18,3C,
3D,CA,72,EA,09,C3,6A,EA,CD,4A,00,F5,01,1
F,00,2B,CD,4A,00,23,CD,4D,00,2B,0B,79,B0
,C2,79,EA,F1,CD,4D,00
2120 DATA 3A,51,C3,01,20,00,21,00,18,3C,
3D,CA,9E,EA,09,C3,96,EA,CD,4A,00,F5,01,1
F,00,23,CD,4A,00,2B,CD,4D,00,23,0B,79,B0
,C2,A5,EA,F1,CD,4D,00,C9
2130 SCREEN 0:COLOR 15,4,4
```

****

C L U B N E W S

A BIG THANK YOU TO 'SMUG OF MONTREAL, CANADA FOR THEIR MUCH APPRECIATED CONTRIBUTIONS TO OUR NEWSLETTER. WE TRUST THEIR CLUB WILL EXPAND AND CONTINUE THE MUCH-NEEDED SUPPORT WHICH SVI AND MSX DESERVES.

****

A BIG WARM WELCOME TO 'SMUG OF CAPETOWN. WHO AFTER SO MUCH PREPARATION, HAVE JUST RELEASED THEIR <INTRO> MAGAZINE AND WHAT A MAGAZINE!
GREENSTAR WISHES STEVEN AND 'SMUG OF CAPETOWN ALL THE BEST FOR THE FUTURE. MAY YOUR CLUB FLOURISH AND CONTINUE SUPPORT FOR SVI, MSX AND CP/M

****

VAAL-TRIANGLE-CLUB.....IMPORTANT NOTICE TO ACTIVE MEMBERS.

KINDLY NOTE THAT THE CLUB WILL MEET ON THE <FIRST> SATURDAY OF EACH MONTH AS FOLLOWS.....
ENGLISH-SPEAKING.....09h00 TO 11h00
AFRIKAANS-SPEAKING.....11h00 TO 13h00
ALL AGES WELCOME AND ALL LEVELS OF ADVANCEMENT ARE CATERED FOR. REMEMBER TO BRING YOUR OWN COMPUTER AND MONITOR / T.V.

****

HOTLINE NUMBERS

WOULD ALL MEMBERS KINDLY TAKE NOTE THAT AS FROM 1986-08-01, THE FOLLOWING HOURS AND NUMBERS WILL APPLY.....

MONDAY TO FRIDAY, 08h00 TO 18h00 and
SATURDAY, 13h00 TO 18h00 and
SUNDAY, 08h00 TO 15h00
(016) 31-9200 (GEORGE VAN DIJK)

MONDAY TO SUNDAY, 18h00 TO 22h00
(016) 6-7125 (BERNARD BURKE)

****

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S W O P - S H O P

SV728/767.....R400.00 F. DU TOIT (011) 28-1717 (X328)
707 D/D + SOFTWARE..R900.00

SV328.....R350.00 F. DU TOIT (011) 28-1717 (X328)
SV605 (S/S 1-DRIVE..R800.00
incl.806/805/PR.CABLE

TAXAN GREEN MONITOR.R320.00 A.LE GRANGE (011)826-5546 (RES)

SV328.....R300.00 J.VAN JAARSVELD (014662)3513 (RES)
TAX.GRN.MONITOR.R300.00 or (01441)22101 (X444)
SV605DM +S/W...R1900.00

CANON V20/JOYSTICK/TAX.COLOUR MONITOR/707 D/D +S/WARE
R2800.00 C.KOTZE (01481) 3484 (RES)
or (01481) 4018 (BUS)

SV605A 2xS/S D/D R800.00 B.BURKE (016) 67125 (RES)

16k RAM + HOMEMADE MINI-EXPANDER
R80.00 F.SKOLTZ (016) 32-3849 (RES)

****

NOTES ON THE MAIL-ORDER LISTS.....

BROTHER M-1009.....SOLD-OUT
MITSUBISHI DISK-DRIVE.....SOLD-OUT
BASIC & MACHINE-CODE COURSE..DELETED FROM THE CATALOGUE DUE TO LACK OF RESPONSE (2 ORDERS IN 3 MONTHS)

****

IN NEXT MONTH'S NEWSLETER.....

NEWS OF SPECTRAVIDEO'S MSX-3.0, MSX AND IBM-COMP. COMPUTER.

****

COMPUTO-LINK UPDATE.....

POWERSOFT HAVE ANNOUNCED THE INTRODUCTION OF A DATABASE TO SUPPORT ALL USERS OF COMPUTO-LINK. THE ADDITIONAL MODULE IS TO BE PROVIDED FREE-OF-CHARGE WHEN IT BECOMES AVAILABLE AND WILL ALLOW USERS TO DOWNLOAD PROGRAMS VIA TELEPHONE-LINE AT REQUEST, FROM A LIST OF PROGRAMS WHICH WILL BE CONTINUALLY EXPANDED. WATCH YOUR NEWSLETTER FOR MORE NEWS OF EXCITING & REWARDING DEVELOPMENTS IN THIS REGARD.

****

NEWS FROM POWERSOFT.....

'POWERBASIC' FOR ALL FORMATS IS COMING UP SHORTLY AND WILL INCORPORATE SOME EXCITING FEATURES SUCH AS.....
VARIABLE REFERENCE (WHERE HAVE I USED VARIABLE SO-AND-SO)
SCREEN-DUMP
'OLD' FOR GETTING BACK THOSE ACCIDENTALLY 'NEWED' PGMS
'DEEK' AND 'DOKE' FOR 16-BIT PEEKS AND POKES
'STACK' FOR THE STACK-POINTER POSITION AT ANY TIME
'STRING' FOR THE STRING-POINTER POSITION
'SIZE' (OF PROGRAM IN BYTES)

AND MANY MORE.....I CAN'T WAIT, CAN YOU ?

****

B E L T E L N E W S

THE DATABASE ON BELTEL CONSISTS OF ONE-HUNDRED PAGES OF INTERESTING INFORMATION FOR SVI, MSX AND CP/M USERS.

NOW IS YOUR CHANCE TO GET HUNDEDS OF PUBLIC-DOMAIN S/WARE.

FOR MORE INFO. ON THIS, DIAL INTO *3077# ON BELTEL NOW!

IF YOU ARE NOT ON BELTEL YET, YOU DON'T KNOW WHAT YOU'RE MISSING !!!!!!

****

VOLTAGE REGULATORS

SECTION 1

BASIC REGULATOR THEORY

A. THE IC VOLTAGE REGULATOR

The basic functional block diagram of an integrated circuit voltage regulator is shown in Figure 1-1. It consists of a stable reference, whose output voltage is V_{REF}, and a high gain error amplifier. The output voltage, V_O, is equal to, or a multiple of, V_{REF}. The regulator will tend to keep V_O constant by sensing any changes in V_O and trying to return it to its original value. Therefore, the ideal voltage regulator could be considered a voltage source with a constant output voltage. However, in practice the IC regulator is better represented by the model shown in Figure 1-2.

In this figure, the regulator is modeled as a voltage source with a positive output impedance, Z_O. The value of the voltage source, V, is not constant; instead, it varies with changes in supply voltage, V_{CC}, and with changes in IC junction temperature, T_J, induced by changes in ambient temperature and power dissipation. Also, the regulator output voltage, V_O, is affected by the voltage drop across Z_O, caused by the output current, I_O. In the following text, the reference and amplifier sections will be described, and their contributions to the changes in the output voltage analyzed.

B. THE VOLTAGE REFERENCE

Naturally, the major requirement for the reference is that it be stable; variations in supply voltage or junction temperature should have little or no effect on the value of the reference voltage, V_{REF}.

The Zener Diode Reference

The simplest form of a voltage reference is shown in Figure 1-3a. It consists of a resistor and a zener diode. The zener voltage, V_Z, is used as the reference voltage. In order to determine V_Z, consider Figure 1-3b. The zener diode, VR1, of Figure 1-3a has been replaced with its equivalent circuit model and the value of V_Z is therefore given by (at a constant junction temperature):

$$V_Z = V_{BZ} + I_Z Z_Z = V_{BZ} + \left(\frac{V_{CC} - V_{BZ}}{R + Z_Z}\right) Z_Z \qquad (1)$$

where V_{BZ} = zener breakdown voltage

I_Z = zener current

Z_Z = zener impedance at I_Z

Note that changes in the supply voltage give rise to changes in the zener current, thereby changing the value of V_Z, the reference voltage.

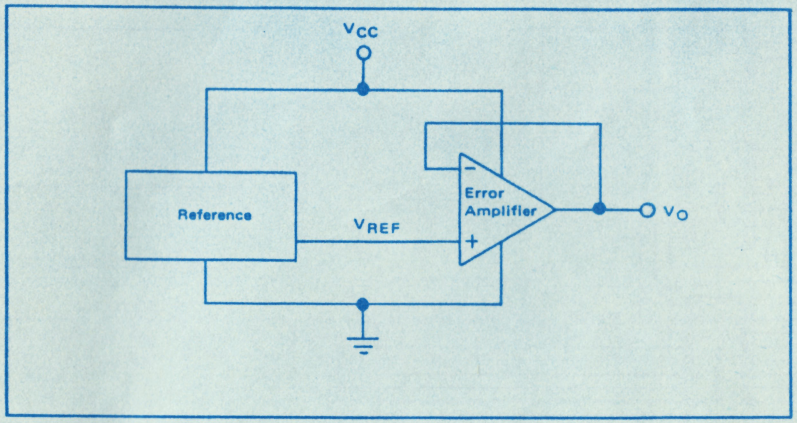


Figure 1-1. Voltage Regulator Functional Block Diagram

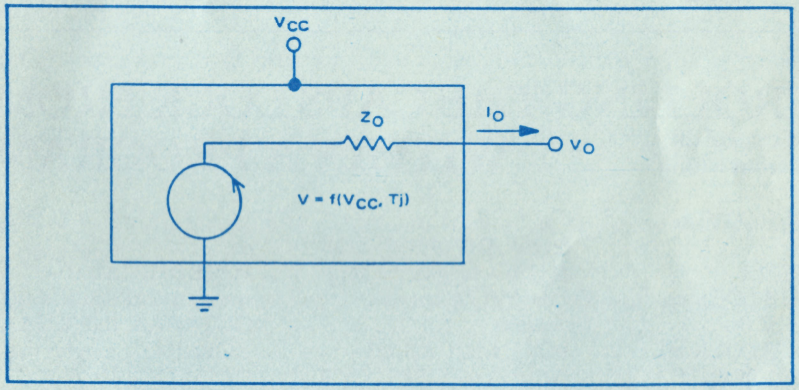


Figure 1-2. Voltage Regulator Equivalent Circuit Model

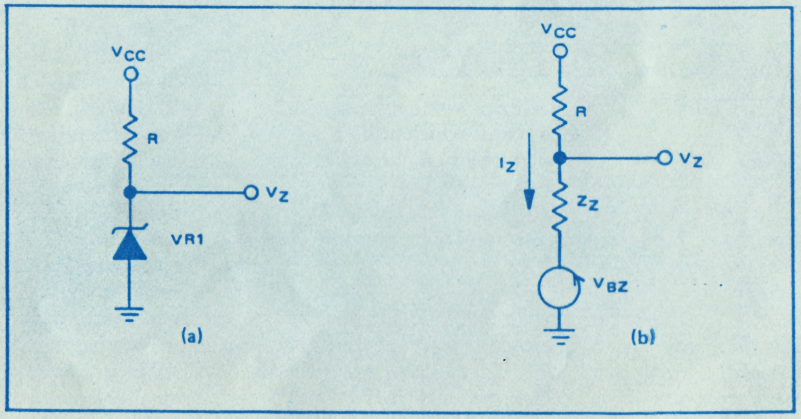
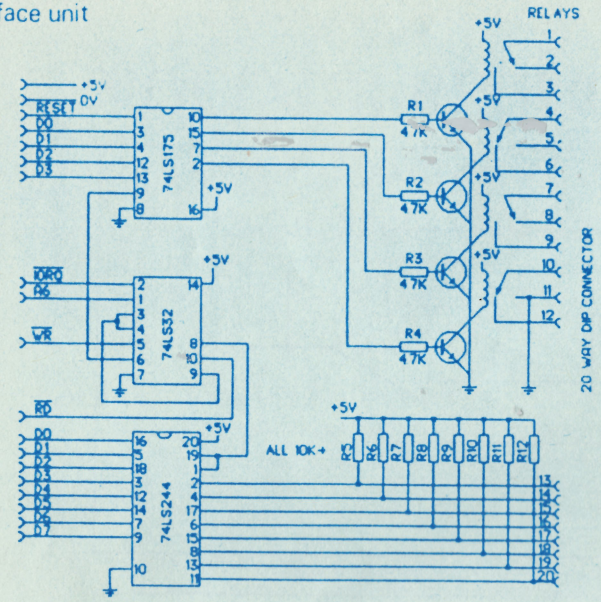
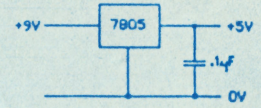


Figure 1-3. Zener Diode Reference

Interface unit





NEXT MONTH'S PROJECT

Transistor

Resistor

Relay Coil

Connection to digital Ground

CONTROLLING THE CAPS LOCK FROM WITHIN A PROGRAM

Q. How can I control the CAPS LOCK from within a program?

A. On the SVI 318/328 use the following:
POKE &HFE38,1 – switches CAPS LOCK on.
POKE &HFE38,0 – switches CAPS LOCK off.
OUT&H88,&H0F:OUT &H8C, INP (&H90) XOR & H20 – toggles the lamp.

On the SVI 728 use the following:
POKE &HFCAB,255 – switches the CAPS LOCK on.
POKE &HFCAB,0 – switches CAPS LOCK off.
OUT&HAA, INP(&HAA) XOR&H40 – toggles the CAPS LOCK lamp.

DIFFERENCES BETWEEN MSC BASIC – SVI BASIC

MSX BASIC QUESTION
Q. Is MSC BASIC different to SVI BASIC?

A. MSX basic differs from SVI basic in a number of small ways. The main differences are given below:

1) Screen modes – MSX has 4 screen modes namely:
i) SCREEN 0 – normal text mode with up to 40 columns.
ii) SCREEN 1 – 32 column text mode which utilises the graphics 1 mode of the video chip. Text is restricted to two colors but sprites are supported.
iii) SCREEN 2 – HI-RES graphics like SVI SCREEN 1 except that printing to SCREEN 2 must be done via a graphics print file.
iv) SCREEN 3 – LO-RES graphics.

2) TAPE SYSTEM – The SVI computers SAVE to tape at 1800 baud rate and the MSX computers use 1200 or 2400 baud. It is therefore not possible to load a MSX program directly into the SVI 318/328.

There are several other differences in the basic but the languages are essentially the same so that the SVI 318/328 programmer will have no problem using MSX.

Say to a pal 'What's this?' Then wave your hand in a snake-like motion. Your pal will say 'I don't know', so then you wave your other hand in a similar manner and say 'Nor do I — but here comes another one!'

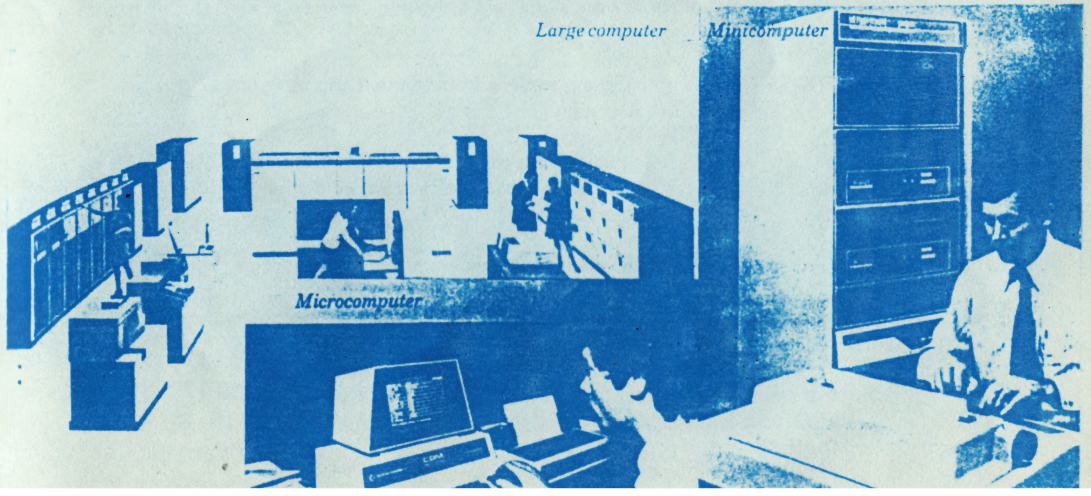
A man committed suicide by hanging himself from a high beam in a remote barn. There was no way to climb up to the beam, and the only thing the police found in the barn was a puddle of water beneath the body. How did he get up to the beam in order to hang himself?
Answer: He stood on a block of ice!

THE PARTS OF A COMPUTER

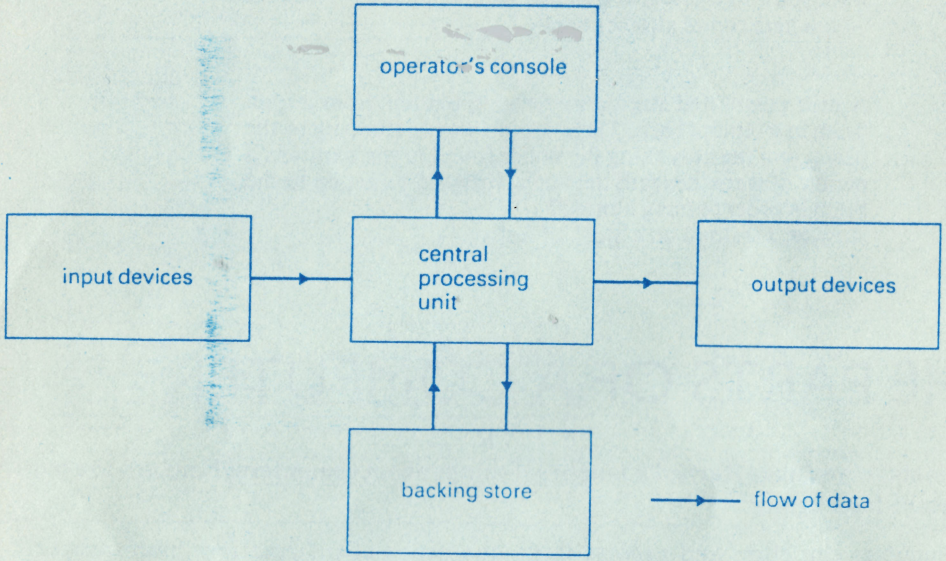
At this stage, you have learned a number of things about computers. Here are the most important facts again:

- Computers can store and process large quantities of information, more properly called **data**.
- Working with data requires three operations: These are **input**, **processing** and **output**.
- Computers work automatically: They are told what to do by sets of instructions called programs.

This chapter introduces the parts of a computer which do these tasks. As you can see from the photographs, computers vary considerably in size. Small computers, sometimes called **microcomputers**, have most of their parts in a single box, or **unit**. Large computers are made up of a number of separate units.



OVERALL STRUCTURE OF A COMPUTER Whether they are separate units, or all in one unit, all computers contain the parts shown in the diagram below:



Block structure of a computer

From the diagram you can see that a computer consists of a number of devices. These devices work together to input, process and output information.

This leads to the important idea of a **system**. A system is a collection of parts working together. People can be parts of a system. Because they consist of a number of devices working together, computers are sometimes called systems.

The components which make up a computer are called **hardware**. Hardware includes the electronic circuits, wires, switches, magnets and other devices found in a computer.

EXERCISE

1 Answer these questions from the text you have just read:
- **a** What is a unit?
- **b** What is a system?
- **c** Do all computers contain a number of units?
- **d** Why is a computer sometimes called a system?
- **e** What is hardware?

2 Some examples of systems are: a motor car, a rugby team, a city. Write down some more examples.

Two boys were born on the same day, on the same date, to the same set of parents. They look alike, talk alike, think alike and behave alike — yet they are not twins! How can this be?

Answer: Because they are two of a set of triplets!

SV-328 DISK BASIC AUTO-LOADER
By: Jean-Guy Fontaine

```
1 CLS:SCREEN0,0:WIDTH39:CLEAR2000:SO$=CHR$(27)+"p":SF$=CHR$(27)+"q"
2 DEFINTA-Z:DIMB$(40),PRG$(40):ZX=DSKF(1)
3 STOPON:ONSTOP GOSUB 32
4 PRINTSO$
5 LOCATE7,0,0:PRINT"                                "
6 LOCATE7,1:PRINT" <<< AUTO RUN PROGRAM >>> "
7 LOCATE7,2:PRINT"     SV 328  DISK BASIC     "
8 LOCATE7,3:PRINT"                                "
9 PRINTSF$
10 DD=1:D$=DSKI$(DD,20,13):DN=ASC(MID$(D$,255,1))
11 GOSUB16:GOSUB26:GOSUB12:RUN
12 LOCATE,,1:PRINT:PRINT"Enter number to run ";:INPUTNN
13 IFNN<1ORNN>I-1THENRUN
14 RUN"1:"+PRG$(NN)
15 CLS:RETURN
16 NB=0
17 FORJ=1TO13:D$=DSKI$(DD,20,J)
18 FORI=1TOLEN(D$)STEP16
19 IFASC(MID$(D$,I,1))=0THEN23
20 IFASC(MID$(D$,I,1))=255THEN24
21 NB=NB+1:B$(NB)=MID$(D$,I,10)
22 B$(NB)=MID$(D$,I,10)
23 NEXTI
24 NEXTJ
25 RETURN
26 CLS:LOCATE0,0:PRINT"Files on disk ";DN;"Free space";ZX:IB=0:IC=1:ID=0
27 FORI=1TONB
28 PRG$(I)=LEFT$(B$(I),6)+"."+MID$(B$(I),7,3)
29 LOCATEIB,IC,0:PRINTI;"> ";PRG$(I)
30 IB=IB+20:IFIB>20THENIB=0:IC=IC+1
31 NEXT
32 RETURN
```

WOORDELYS

BEHEERVERANDERLIKE	CONTROL VARIABLE
DUBBELNOUKEURIG	DOUBLE PRECISION
EINDIGE LUS	FINITE LOOP
FOUTBOODSKAP	ERROR MESSAGE
GESLOTE LUS	INFINITE or CLOSED LOOP
GENESTE LUS	NESTED LOOP
HEELTAL	INTEGER
LUS	LOOP
NUMERIESE VERANDERLIKE	NUMERIC VARIABLE
OOP LUS	OPEN or FINITE LOOP
SINTAKSIS	SYNTAX
STELLING	STATEMENT
TERUGKRING	LOOP BACK
UITDRUKKING	EXPRESSION
UITVOERING	EXECUTION
VERANDERLIKE	VARIABLE
VERSTEKWAARDE	DEFAULT VALUE
VERTRAAGLUS	DELAY LOOP

MSX 2.Ø AND MITSUBISHI

By: Frances Fontaine

The MSX computer companies are currently releasing in Japan and Europe an upgraded version of MSX called MSX 2.Ø.

At the present time we Canadians are able to purchase only MSX plus computers, which are versions of MSX 1.[illegible] extras added to the standard, such as the CX5M's musical capabilities and the SV-738's CP/M, incorporated disk drive, 8Ø column, RS232 interface and centronic interface.

Mitsubishi along with other MSX companies will be releasing the newer version of MSX which uses a new video chip the V-9938 which will allow a maximum resolution of 512 x 424 with 16 colours on screen. This new chip will allow you to combine external signals from video cameras, VCRs and television with computer generated images. This effect is known as super imposing images and there are new BASIC commands added to MSX 2.Ø which allows you to control the super impose features.

The new sound generator in the MSX 2.Ø computers will be similar to the sound chip the CX5M uses.

They have kept the Z8ØA chip as the CPU to allow for upward compatibility between MSX 1.Ø and MSX 2.Ø. Thus allowing you to use your software that you originally purchased for your MSX 1.Ø, if you should decide to upgrade to an MSX 2.Ø in the future. Perhaps in future upgrades of MSX like MSX 3.Ø we may start seeing 16 bit or 32 bit chips along with a Z8ØA co-proccessor being used as the CPU. The Z8ØA will be added to maintain compatibility with the first generations of MSX.

Mitsubishi has added a numeric keypad to it's MSX 2.Ø computer and they have also produced a digitiser that may be used with the computer to create amazing video/graphic effects. The digitiser is used to capture signals from sources such as a VCR or video camera, to be processed by the computer. Thus you are able to capture and freeze images, then you could edit them as you like. You may super impose other images on top or beside the first one, or you may cut out, paint, reduce, enlarge, compress or stretch your image or any portion of it, as you'd like. Then you may combine the final image with a "live" image from the TV camera using the super impose feature. You may also save your image to tape or disk or make a print out of it.

Some of MSX 2.Ø's new commands are: SET SCREEN, COLOR (C), COLOR=NEW, COLOR=RESTORE, COLOR SPRITES(P)=, COLOR SPRITE(P)=V, COPY, SET VIDEO, COPY SCREEN (M), GET DATE X$, GET TIME X$, SET DATE, SET TIME, SET ADJUST(X,Y), SET BEEP, SET TITLE, SET PASSWORD and SET PROMPT.

MSX 2.Ø may also be able to use CP/M and the large CP/M program library. Thus the MSX 2.Ø computers will be very useful and versatile. Allowing it to be used for a variety of applications, from game playing, musical and video capabilities to business purposes.

The only question we have is when will MSX 2.Ø be available here

One thing that consumers should be aware of though, is that if any MSX computer does not use those commands and have the following specifications, then it's not a real MSX 2.Ø only an MSX 1.Ø in disquise. The following is a list of the MSX 2.Ø's specifications:

CPU - Z8ØA

MEMORY - 48k ROM
- 128k VRAM maximum
- 64k RAM minimum
- Battery backed-up RAM

VIDEO DISPLAY - 4Ø x 24 text mode
- 8Ø x 24 text mode
- 32 x 24 text mode
- 256 x 192 graphics (16 colours)
- 64 x 48 graphics (16 colours)
- 256 x 212 bitmapped graphics (16 out of 512 colours)
- 512 x 212 bitmapped graphics (4 out of 512 colours)
- 512 x 212 bitmapped graphics (16 out of 512 colours)
- 256 x 212 bitmapped graphics (256 colours)
- Maximum of 512 colours available
- Up to 8 sprites per line
- Super impose features

OPTIONAL FEATURES - Light Pen
- Kanji Character Generator
- Audio Visual Control
- Mouse
- 9 Voice FM Synthesis sound chip

QUADRATIC EQUATION SOLVER

One of the most familiar equations to engineers and scientists is the quadratic equation. This program solves quadratic equations for all three special cases: real roots, real repeating roots, and complex roots. It solves equations of the form

$$A X\uparrow 2 + B X + C = Ø$$

and simply requires that the coefficients of the equation A, B, and C be entered.

```
1  PRINT : PRINT : PRINT : PRINT "WRITTEN BY JULES H. GILDER"
2  PRINT : PRINT
3  FOR X = 1 TO 4000: NEXT X
4  PRINT : PRINT : PRINT : PRINT
5  PRINT  TAB( 7)"QUADRATIC EQUATION SOLVER"
6  PRINT : PRINT : PRINT : PRINT
40  PRINT "THIS PROGRAM SOLVES QUADRATIC EQUATIONS"
50  PRINT "OF THE FORM:"
60  PRINT "              AX↑2+BX+C=0"
70  PRINT
80  PRINT "FOR REAL, COMPLEX AND REPEATING ROOTS."
90  PRINT
100  INPUT "ENTER COEFFICIENTS A,B,C: ";A,B,C
110  PRINT
115  PRINT : PRINT
120 Z = B ↑ 2 - (4 * A * C)
125 Z =  INT (100 * Z + .5) / 100
130  IF  Z < 0  THEN  200
140  IF  Z = 0  THEN  180
150 X1 = ( - B +  SQR (Z)) / (2 * A)
155 X1 =  INT (100 * X1 + .5) / 100
160 X2 = ( - B -  SQR (Z)) / (2 * A)
165 X2 =  INT (100 * X2 + .5) / 100
170  PRINT : PRINT "THE ROOTS TO THE EQUATION ARE REAL"
172  PRINT
174  PRINT "ROOT 1 = ";X1;"  ROOT 2 = ";X2
175  END
180 X1 =  - B / (2 * A)
185 X1 =  INT (100 * X1 + .5) / 100
190 X2 = X1
192  PRINT
193  PRINT "THE ROOTS TO THE EQUATION ARE REAL AND"
194  PRINT "REPEATING."
195  GOTO  172
200 R1 =  - B / (2 * A)
205 R1 =  INT (100 * R1 + .5) / 100
210 R2 = R1
220 I1 =  SQR ( ABS (Z)) / (2 * A)
225 I1 =  INT (100 * I1 + .5) / 100
230 I2 =  - I1
240  PRINT
250  PRINT "THE ROOTS TO THE EQUATION ARE COMPLEX"
260  PRINT
270  PRINT "ROOT 1 = ";R1;" + J ";ABS (I1)
280  PRINT "ROOT 2 = ";R2;" - J ";ABS (I2)
290  END
```

****

CROSSWORD SOLUTION

ACROSS – 1. Word-formation; 9. Tosspot; 10. Painful; 11. Custom; 12. Algernon; 14. On the house; 16. Film; 19. Turn; 20. Edward Lear; 22. Clubroom; 23. Jumble; 26. Primers; 27. Outrage; 28. General strike.

DOWN – 2. Oasis; 3. Deplored; 4. Oats; 5. Twined; 6. Offensive; 7. Stick out; 8. Glen; 10. Pole Star; 13. Wondrous; 15. Turquoise; 17. Marketed; 18. Adjuster; 21. Greece; 22. Copy; 24. Blank; 25. Boos.

Lucky Dip

'Did you hear about the fool who keeps going round saying No?'
'No.'
'Oh, so it's you!'

Boolean Algebra for Computer Logic

Operations

Although Boolean algebra has many things in common with conventional algebra, it does have differences. Although the two possible values of 0 and 1 do not necessarily represent numbers, let's temporarily consider them as such. If you take the product $1 \cdot 1 \cdot 1 \cdot 1$ (one times one times one times one) you have a product of 1. This is said to equal 1, and the = sign has conventional meaning of equality.

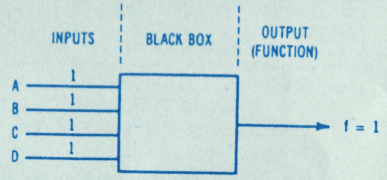


Fig. 1-1. If A and B and C and D are all 1s, the function f (output) is a 1.

A computer logic element is a "black box" which has two or more inputs, with a single output that is a direct function of the inputs. Fig. 1-1 shows such a box with four inputs, all of which are "highs" or logical ones. If the function (output) is a 1, all we know at the present time is that when all inputs are high, the function (f) is also high (an output occurs).

Now consider the product $1 \cdot 0 \cdot 1 \cdot 1$. You will recall that 1 times 0 is always 0, and 0 times 1 is always 0. Thus the product of $1 \cdot 0 \cdot 1 \cdot 1 = 0$ (no output occurs).

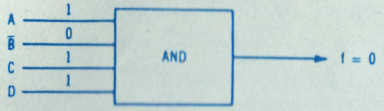


Fig. 1-2. If any one or all inputs are at logic 0, then f = 0 (no output occurs).

Fig. 1-2 shows the black box of Fig. 1-1 with a 0 on the B input ($\bar{B}$) and reveals that the output (f) is now 0. Further tests reveal that if *any one* of the inputs is a 0, the function (output) is also 0. In symbolic logic, this is termed an AND circuit. All inputs must be simultaneously high for an output to occur. If any one input is 0, or if all inputs are 0, the output will be 0. Note that this says A AND B AND C AND D (written ABCD) must all be high (no bars) for a "1" output to occur. In Fig. 1-2, $A\bar{B}CD$ (read A AND B NOT AND C AND D) = 0. Note, therefore, that the multiplication sign in Boolean algebra is Boolean operation AND. Usually, as in conventional algebra, the multiplication (times) sign is simply deleted.

Now let's consider the sum (add) operation. Suppose you have 1+1+1+1. In conventional arithmetic, this would equal the quantity 4. The conventional number system has a base of ten, and consists of numbers 0–9. When you reach the sum of 9, you change that column to 0 and start a second column (to the left) with a 1, to obtain the quantity 10 (ten).

A number system that consists of only 0 and 1 is termed a *binary* (base 2) system. Since 1 is the highest number in the system, 1+1 = 0 and a 1 is carried to the left as: 10. This does *not* read "ten," but it is read as "one-zero." You will get acquainted with the binary system in this chapter.

Assume for the moment you have the black box of Fig. 1-3. You find that if either or both inputs are high, an output occurs. This says that if A or B (or A or B or C or D, etc.) is high, an output occurs. Since an output occurs if either one or all inputs

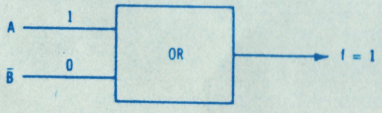


Fig. 1-3. If a 1 exists at any input, the function f (output) is a 1.

are high, it is termed an *inclusive* OR Boolean operation. In Boolean notation, the connective OR is signified by the sum (+) sign. Thus A + B + C + D reads A OR B OR C OR D. This says that an output will occur (f = 1) if any one or more inputs is 1. If all inputs are 0, no output occurs (f = 0).

How a combination of AND and OR black boxes is used to obtain binary sums and other mathematical as well as logic statement operations will be revealed after you learn the technique of the binary system. To summarize thus far:

AB means A AND B.
A + B means A OR B.
$\overline{AB}$ means NOT A AND B.

A bar over a letter or group of letters always indicates the inverted form of the indicated letter or quantity.

1-2. THE POWER OF A NUMBER (EXPONENT)

In our conventional decimal system, $10^1 = 10$ and $10^{-1} = 0.1$. Now $10^0 = 10^{1-1} = 10^1 \times 10^{-1} = 10 \times 0.1 = 1$. Thus $10^0 = 1$. *Any* number raised to the zeroth power is equal to 1. So $2^1 = 2$ and $2^{-1} = 0.5$ and $2^0 = 1$.

Table 1-1 reviews the decimal (base 10) system for four digits.

Table 1-1. Decimal 1 System for Four Digits

	Positional Weight			
	10^3	10^2	10^1	10^0
Equivalent	1000	100	10	1
3248 =	3	0	0	0
		2	0	0
			4	0
				8
	= 3 (MSD)	2	4	8 (LSD)
Showing That 3248 = 3000+200+40+8				

COMBINATIONS

In the permutations program, the number of ways n objects could be arranged in order into groups of r was calculated. Sometimes the order of the objects in a group is not important—for example, the number of ways 12 people can be divided into groups of 3. Another application is found in error detection and correction codes for communication systems.

The number of ways items can be grouped into groups of r objects, where order is not important, is called the "combination."

The combination of 12 objects taken 2 at a time is equal to

$$C = 12!/(3!(12 - 3)!) = 220$$

The generalized formula for calculating combinations is

$$C = n!/(r!(n - r)!)$$

The difference between this calculation and the one for permutations is the extra r! term in the denominator.

Once again, the factorial calculation is carried out by a subroutine located at line 220. The results of three calls to this subroutine are used to calculate the combinations. As in the permutations program, error detection tests are included.

```
1  PRINT : PRINT : PRINT : PRINT "WRITTEN BY JULES H. GILDER"
2  PRINT : PRINT
3  FOR X = 1 TO 4000: NEXT X
10  PRINT : PRINT : PRINT : PRINT
20  PRINT "                COMBINATIONS"
30  PRINT : PRINT : PRINT : PRINT
40  PRINT "THIS PROGRAM COMPUTES THE COMBINATIONS"
50  PRINT "OF 'N' OBJECTS TAKEN 'R' AT A TIME."
60  PRINT : PRINT : PRINT
70  INPUT "ENTER NUMBER OF OBJECTS.  ";N
80  INPUT "ENTER NUMBER PER GROUP.  ";R
90  PRINT : PRINT : PRINT
100  IF  N < R  THEN  PRINT "INPUT ERROR - N<R": GOTO  60
110  IF  R < 0  THEN  PRINT "INPUT ERROR - R<0": GOTO  60
120 X = N
130  GOSUB  220
140 NN = XF
150 X = N - R
160  GOSUB  220
170 NR = XF
172 X = R
174  GOSUB  220
176 RR = XF
180 C =  INT (NN / NR / RR + .5)
185  PRINT : PRINT : PRINT
190  PRINT " THERE ARE ";C;" COMBINATIONS OF"
195  PRINT
200  PRINT N;" OBJECTS TAKEN ";R;" AT A TIME."
210  GOTO 270
220 XF = 1
230  FOR  K = 1  TO  X
240 XF = K * XF
250  NEXT  K
260  RETURN
270  END
```

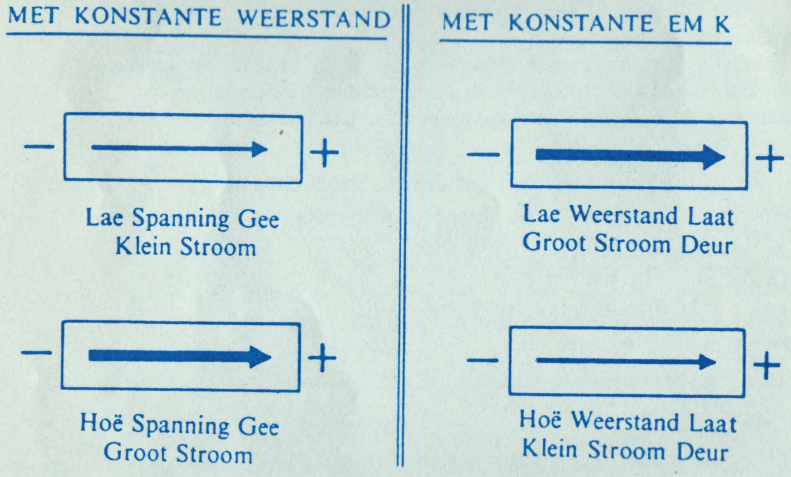
> YOU CAN never go wrong by giving a youngster lots of love and kisses mixed with discipline. Child training is merely knowing which end of your child to pat . . . and when.
>
> —Jan Marshall, *Still Hanging In There*

§2: DIE WET VAN OHM

U het alreeds in Deel 1 geleer dat daar 'n vaste verband bestaan tussen die **em k** wat 'n stroom elektrone deur 'n kring aandryf, die **weerstand** van daardie kring, en die tempo van die **stroomvloei** deur die kring.

Met 'n konstante weerstand in 'n kring, het u geleer dat stroomvloei sal **toeneem** namate die em k, wat op die kringe aangelê word, toeneem. Met 'n konstante em k op die kring aangelê, sal die stroomvloei afneem namate die weerstand van die kring toeneem. Hierdie twee stellings kan u soos volg kombineer: *Stroomvloei in 'n kring neem toe namate die em k verhoog word, en neem af namate die weerstand verhoog word.*

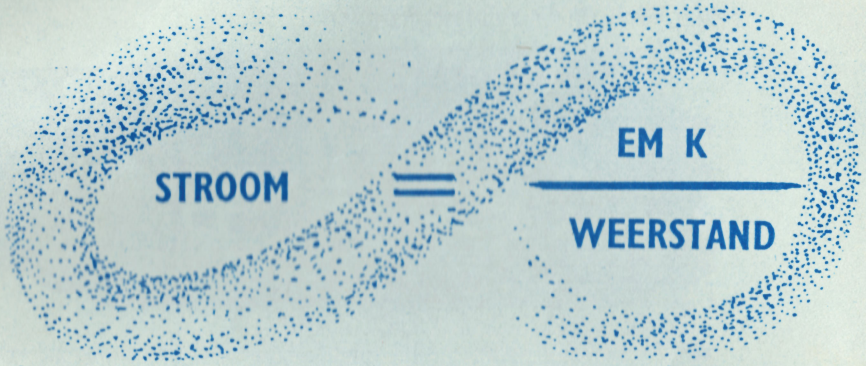
ELEKTRONESTROOMVLOEI



U het ook geleer dat die Duitse Fisikus, Georg Simon Ohm, 'n studie gemaak het van die verband tussen em k, stroom en weerstand. Sy stelling van hierdie verband is een van die fundamentele wette van die fisika. Dwarsdeur u werk met elektrisiteit, sal u gedurig van hierdie Wet gebruik maak, asook later, of u van voorneme is om 'n studie te maak van elektriese masjinerie, elektronika, radar, rekenaars, servomeganismes—of, in werklikheid enigiets waarby die vloei van 'n elektriese stroom betrokke mag wees. Wat dan is die stelling van hierdie deurslaggewende Wet? Een van die mees algemene maniere om dit te stel, is soos volg:

> **Die stroom, wat deur 'n kring vloei, is *Direk* eweredig aan die aangelegde em k en *Omgekeerd* eweredig aan die weerstand.**

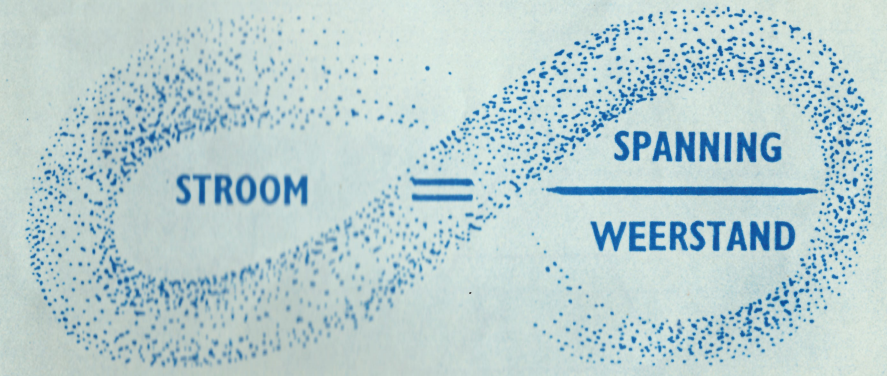
Dit is moontlik om die Wet van Ohm, soos onderaan die vorige bladsy uiteengesit, as 'n wiskundige vergelyking uit te druk, soos volg:



Hierdie vergelyking moet u leer en nooit weer vergeet nie. *Dit is die belangrikste feit tot dusver, in hierdie reeks, wat aan u meegedeel is.*

Met hierdie vergelyking nou stewig in u gedagte ingeprent, gaan nou terug na iets wat u in Deel 1 geleer het. U sal onthou dat u meegedeel is dat sowel elektromotoriese krag as potensiaalverskil in volt gemeet word. As 'n elektrotegniese ingenieur 'n meting in 'n kring met 'n voltmeter doen, is hy nie eintlik besorgd daaroor om te weet of dit 'n em k of 'n pv. is nie, want die een is in werklikheid maar 'n weerspieëling van die ander. Die lesing wat hy kry, noem hy maar 'n „spanning". Daarom, as hy bepaal dat 'n pv. tussen twee punte in 'n kring bestaan, is die ingenieur geneig om te sê dat „'n spanning tussen die punte bestaan" of dat daar „*'n spaaningsval oor die twee punte is*". Met ander woorde—hy is geneig om die woord „spanning" te gebruik om òf 'n elektromotoriese krag òf 'n potensiaalverskil voor te stel, behalwe as daar 'n spesifieke rede is waarom daar tussen die twee onderskei moet word.

As hierdie manier om sake te stel, aanvaar word, (*soos voortaan in hierdie reeks die geval sal wees*) kan Ohm se wet soos volg voorgestel word:



Die Wet van Ohm (*vervolg*)

By elektriese notasie word stroom altyd voorgestel deur die letter „I", weerstand deur die letter „R" en spanning deur die letter „V".

U kan dus die stelling van die Wet van Ohm soos dit onderaan die vorige bladsy staan, op die volgende wyse skryf:

$$I = \frac{V}{R}$$

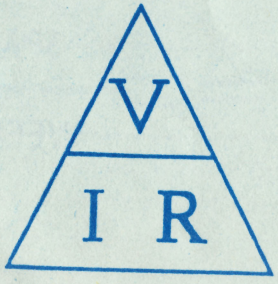
Met behulp van die eenvoudigste algebra kan hierdie belangrike vergelyking ook soos volg geskryf word:

$$V = I \times R \quad \text{of} \quad R = \frac{V}{I}$$

Welke van die drie maniere om die Wet van Ohm te skryf, gebruik sal word, hang van twee dinge af. Eerstens, watter feite het u tot u beskikking aangaande die kring, wat u besig is om te oorweeg? Tweedens, watter feite wil u nou vasstel?

Daar is gelukkig 'n maklike manier waardeur u kan onthou watter formule gebruik moet word. Noem dit, as u wil, die „towerdriehoek".

Teken 'n driehoek en trek dan 'n horisontale lyn daardeur ongeveer halfpad bokant die basislyn. Skryf die letter „V" in die klein driehoek wat nou bo in die towerdriehoek gevorm is, en die letters „I" en „R" onderkant die lyn soos hieronder aangetoon:



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